

Best Fit Interpolator Plugin

User Tutorial and Practical Guide

A QGIS plugin for spatial mapping, interpolation method comparison, validation, and framework-based method recommendation.

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Laura Delgado Bejarano
M.Sc. Agricultural Engineering – Digital Agriculture
School of Agricultural Engineering – University of Campinas
Campinas, Brazil

GiTAP

Research group

QGIS

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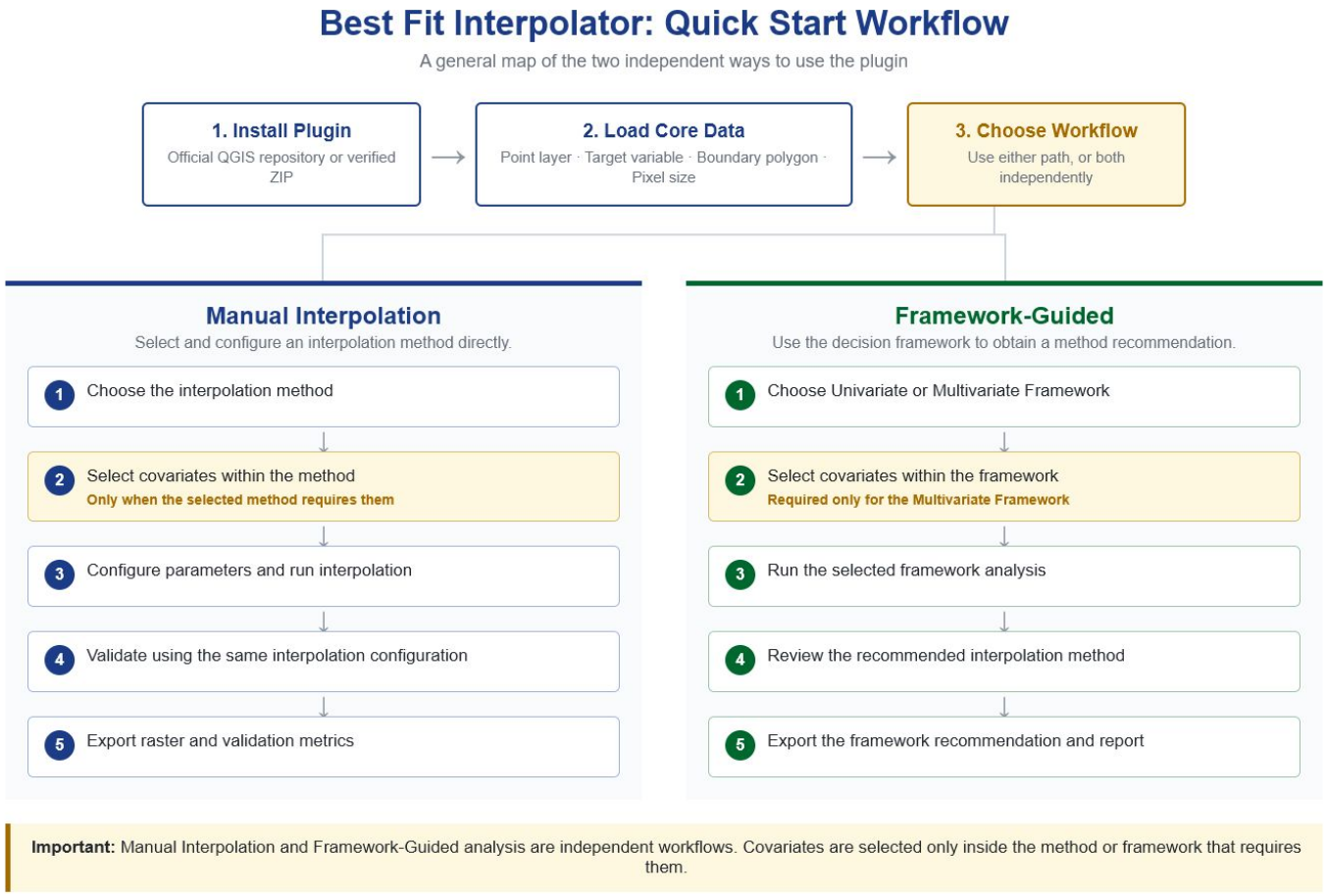


Figure 1: General workflow of the Best Fit Interpolator plugin. Manual Interpolation and Framework-Guided analysis are independent paths.

1 Motivation

Spatial mapping is a fundamental step in many environmental, agricultural, and geospatial applications. Maps are used to represent the spatial distribution of variables, support decision-making, identify management zones, guide field operations, and communicate patterns that cannot be easily interpreted from point-based observations alone. However, producing a reliable map from sampled data is not a purely automatic task. The quality of the final map depends on several factors, including the number of samples, the spatial distribution of the sampling points, the spatial structure of the target variable, the interpolation method, and the validation strategy used to evaluate prediction performance.

In many practical situations, interpolation methods are selected based on habit, software availability, or visual appearance of the output map. This can lead to inconsistent results, especially when the characteristics of the dataset are not considered. A method that performs well for one variable, sampling density, or spatial pattern may not be suitable for another.

The *Best Fit Interpolator Plugin* was developed to support a more transparent, reproducible, and data-driven interpolation workflow in QGIS. The plugin allows users to organize spatial input data, evaluate spatial diagnostics, apply different interpolation methods, compare validation metrics, and select a suitable method according to the characteristics of the dataset. Although the research framework behind the plugin was developed with soil data, the general problem addressed by the plugin is broader: how to choose an appropriate interpolation method for spatial mapping based on sample size, spatial pattern, spatial dependence, and prediction performance. Soil data were used as a case study because soil sampling is often limited by cost, making sampling density a critical factor in the mapping process.

This tutorial provides a step-by-step guide for using the plugin, from data preparation to final map export and report generation.

2 What the Plugin Does

The *Best Fit Interpolator Plugin* integrates tools for spatial interpolation, validation, and method recommendation within QGIS. It was designed to help users perform a complete mapping workflow using point-based observations, boundary polygons, and, when available, auxiliary raster covariates.

The plugin can be applied to spatial mapping problems where the objective is to estimate a continuous surface from sampled observations. Although the framework was developed using soil attributes as the main case study, the plugin structure can support broader interpolation workflows involving different types of spatial variables.

The main functionalities include:

- Selection of point-based observations, target variable, and boundary polygon;
- Definition of output pixel size;
- Global Moran's I, Local Moran / LISA, statistical outlier diagnostics, and explicit sample review;
- Deterministic interpolation methods, such as IDW and Thin Plate Spline;
- Geostatistical interpolation methods, including kriging-based approaches;
- Machine learning interpolation using auxiliary covariates;
- Validation and comparison of interpolation methods;
- Framework-based recommendation according to sample size, spatial pattern, and spatial dependence;

- Side-by-side maps and a signed difference map on a shared grid;
- Interactive session reports, temporary HTML previews, and PDF export;
- Automatic processing profiles for dense and massive datasets.

3 Scientific Background

The framework implemented in the *Best Fit Interpolator Plugin* was developed based on the study by DELGADO BEJARANO et al. (2026), published in *Precision Agriculture*. The study evaluated how data characteristics, including sample size, sampling density, spatial structure, and spatial dependence, influence the performance and selection of interpolation methods for digital soil mapping. The scientific development of the framework was focused on soil mapping because soil sampling campaigns are often constrained by operational and economic costs. As a result, soil datasets commonly present different sampling densities, which directly affect interpolation performance and the uncertainty associated with the final map.

Despite this soil-based research context, the conceptual basis of the plugin is not restricted to soil fertility mapping. The central objective is to support users in understanding how the characteristics of their spatial data influence interpolation performance and method suitability.

The framework module uses spatial indicators, such as Moran's I, spatial pattern classification, and the spatial dependence index, together with validation metrics, to guide the recommendation of suitable interpolation methods. The original study proposed two decision frameworks: a univariate framework (Figure 2), based mainly on the target variable and its spatial structure, and a multivariate framework (Figure 3), which also considers the use of auxiliary covariates.

Information

Article citation

The framework-based recommendation module implemented in the *Best Fit Interpolator Plugin* is based on the following article:

DELGADO BEJARANO, L.; OLIVEIRA, A. L. G.; POZZUTO, J. V. F.; SÁNCHEZ, D. C.; AMARAL, L. R. do. Performance of interpolation methods in digital soil mapping: the influence of data characteristics. **Precision Agriculture**, v. 27, n. 1, article 10, 2026. DOI: https://doi.org/10.1007/s11119-025-10311-8.

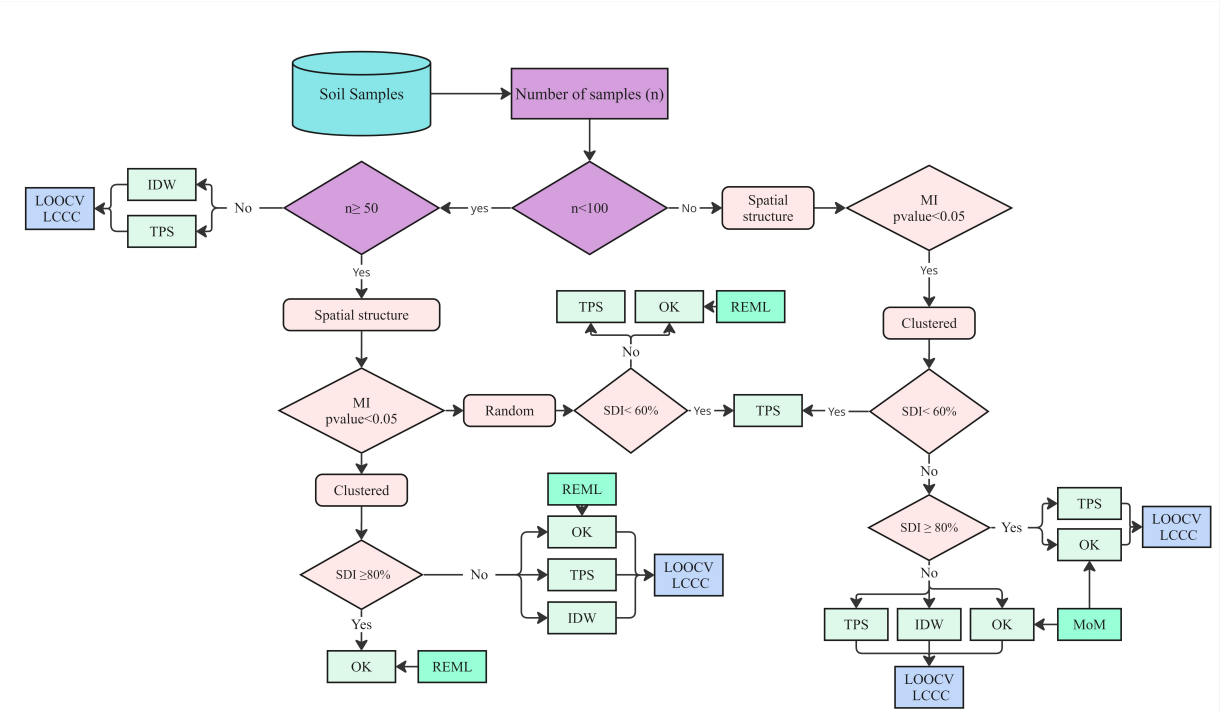


Figure 2: Univariate decision framework for interpolation method selection.

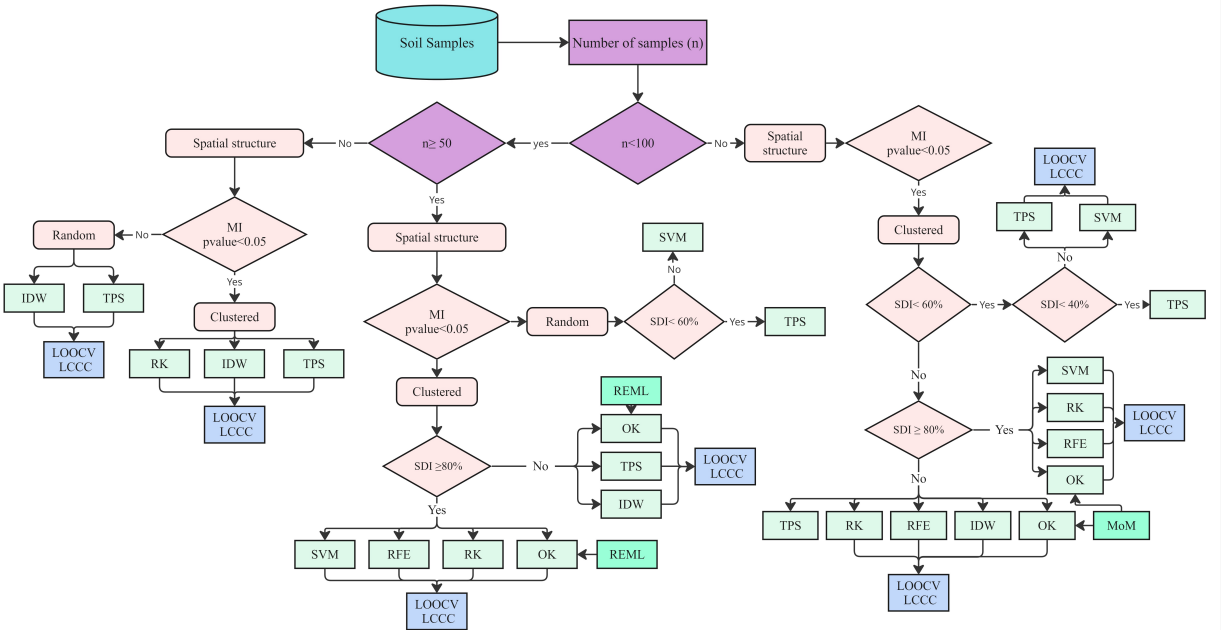


Figure 3: Multivariate decision framework for interpolation method selection using auxiliary covariates.

4 Software Requirements

The minimum declared QGIS version is 3.14. The updated plugin supports the Qt 5 environment used by QGIS 3 and the Qt 6 environment used by QGIS 4. The same workflow is used across these environments; operating-system fonts, window borders, and individual QGIS controls may differ.

This manual describes plugin version 2.1, whose current workflows have been checked on Windows with QGIS 3.44.8 and 4.2.3. Declared compatibility begins at QGIS 3.14. Use Python packages compatible with the Python version and architecture of the installed QGIS.

Table 1: Software requirements and documentation scope.

Component	Description
GIS software	QGIS 3.14 or later; QGIS 3 and 4 interfaces are supported
Python	Use the Python environment distributed with the installed QGIS
Machine learning	A compatible NumPy, SciPy, pandas, and scikit-learn stack in that QGIS environment
REML	The current implementation provides a Python/SciPy backend; a separate R installation is not a general prerequisite
Documentation	User guide for the current interface and analysis workflows

Information

This guide describes the v2.1 controls. Some screenshots illustrate the earlier layout and may not show every added selector; follow the current control names and instructions. Older packaged releases may not include these features. Record the installed plugin version from `About`, the QGIS version, and the operating system for reproducible work.

5 Input Data Requirements

The *Best Fit Interpolator Plugin* requires spatial input data to perform interpolation, validation, and method recommendation. The minimum input data include a point layer containing sampled observations and a polygon layer defining the mapping area. Additional raster covariates can be used for multivariate interpolation methods.

The plugin was developed to support spatial mapping workflows based on point observations. These observations may represent soil attributes, environmental measurements, crop indicators, sensor-based variables, or other spatially distributed variables collected at known locations.

5.1 Point Layer

The point layer contains the sampled observations used as input for interpolation. Each point must have a valid geometry and an attribute table containing at least one numerical variable to be mapped.

Table 2: Minimum requirements for the point layer.

Requirement	Description
Geometry type	Point layer
Coordinate reference system	Projected CRS recommended, preferably UTM
Attribute table	At least one numerical target variable
Missing values	Missing or invalid values should be removed before interpolation
Sampling locations	Points should be located inside or near the mapping boundary

Information

The plugin uses the selected numerical field as the target variable for interpolation. Non-numerical fields should not be selected as interpolation variables.

5.2 Boundary Polygon

The boundary polygon defines the spatial extent of the interpolation. The interpolated raster is generated according to this area and should represent the region where spatial prediction is required.

Table 3: Minimum requirements for the boundary polygon.

Requirement	Description
Geometry type	Polygon layer
Purpose	Defines the interpolation extent and output raster mask
Coordinate reference system	Same CRS as the point layer is recommended
Geometry validity	The polygon should be valid and free of topology errors

Information

The boundary polygon is used to limit the spatial prediction area. It helps avoid generating maps outside the area of interest.

5.3 Auxiliary Raster Covariates

Auxiliary raster covariates are optional input layers that can be used in multivariate interpolation methods. These covariates should represent spatial information that may help explain the variability of the target variable.

Examples of auxiliary covariates include:

- elevation or terrain attributes;
- apparent electrical conductivity;
- vegetation indices;
- bare soil indices;
- remote sensing products;
- proximal sensing layers;
- management or environmental layers.

Information

Auxiliary covariates are especially useful when they are spatially continuous and related to the target variable. Poorly related covariates may not improve interpolation performance.

5.4 Coordinate Reference System

Distance-based interpolation methods require reliable distance calculations. For this reason, the use of a projected coordinate reference system is strongly recommended.

Geographic coordinate systems, such as latitude and longitude, should be avoided for interpolation because distances are expressed in degrees rather than meters.

Table 4: Recommended coordinate reference system configuration.

Layer	Recommendation
Point layer	Projected CRS, preferably UTM
Boundary polygon	Same CRS as the point layer
Raster covariates	Same CRS and compatible resolution when possible
QGIS project	Same projected CRS used by the input layers

Information

Using different coordinate reference systems among point, polygon, and raster layers may cause misalignment, incorrect distance calculations, or failed interpolation outputs.

6 General Workflow

The *Best Fit Interpolator Plugin* follows a shared preparation stage and then separates into two independent analysis paths. The user may run a manual interpolation workflow, choosing one or more deterministic, geostatistical, or machine learning methods, or may use the framework-guided workflow to obtain a method recommendation from the decision framework. These paths are not sequential requirements; the framework does not need to be run after the manual workflow, and the manual workflow does not need to be run before the framework.

6.1 Shared Preparation

Before choosing the analysis path, prepare the project and load the common inputs:

1. Load the input layers in QGIS.
2. Open the *Best Fit Interpolator Plugin*.
3. Select the point layer containing the sampled observations.
4. Select the target variable to be interpolated.
5. Select the boundary polygon that defines the mapping area.
6. Define the output pixel size.
7. Review the spatial diagnostics and the data preview.

6.2 Path A: Manual Interpolation

Use this path when the user wants to choose and run a specific interpolation method directly.

1. Select the interpolation family: deterministic, geostatistical, or machine learning.
2. Configure the selected method and its parameters.
3. For machine learning methods, load and prepare the covariate rasters before fitting the model.
4. Run the interpolation.
5. Evaluate the result using the validation metrics and diagnostic plots.
6. Export the final raster and summary outputs.

6.3 Path B: Framework-Guided Recommendation

Use this path when the user wants the plugin to support method selection using the decision framework. The framework uses the characteristics of the loaded data and the validation information available in the project to guide the recommendation. It is an independent workflow and should be interpreted as decision support rather than as a mandatory step after manual interpolation.

1. Open the framework workflow after loading the common data inputs.
2. Review the data characteristics and diagnostic information used by the framework.

3. Generate the framework-based recommendation.
4. Use the recommendation to decide which interpolation method should be run or reported.
5. Export the framework summary when documentation of the decision process is required.

Information

The plugin does not assume that one interpolation method is universally superior. Manual interpolation and framework-guided recommendation are complementary but independent workflows. The user may choose either path according to the objective of the mapping process.

7 Installing the Plugin

The *Best Fit Interpolator Plugin* can be installed directly from the official QGIS Plugin Repository or from the ZIP package published on GitHub. Installation through the official repository is recommended for most users because QGIS can detect future updates automatically.

Information

Use the official QGIS repository whenever the plugin is publicly available. Use the GitHub ZIP method for a specific release, testing, or controlled installation. Do not extract the ZIP file before installing it in QGIS.

7.1 Installation from the Official QGIS Plugin Repository

Step 1: Open the QGIS Plugin Manager

Start QGIS and open **Plugins** > **Manage and Install Plugins**. QGIS will display the plugin manager and refresh the configured repositories.

Step 2: Search for the plugin

Select **All** in the left panel and type **Best Fit Interpolator** in the search field. Select the plugin when it appears in the results.

Step 3: Install the plugin

Review the plugin name, version, author, and description. Click **Install Plugin** and wait until QGIS confirms that the installation was completed successfully.

Step 4: Verify the installation

Open the **Installed** tab and confirm that **Best Fit Interpolator** is listed and enabled. The plugin can then be opened from the QGIS plugin interface.

7.2 Installation from a GitHub Release ZIP

The verified release packages are published on the Best Fit Interpolator GitHub Releases page. Select the packaged plugin ZIP from the release assets. Its release identifier may change; the installation procedure and this manual's filename remain the same.

Step 5: Download the release package

Open the GitHub Releases page, select the appropriate current release, and download the packaged plugin ZIP from the release assets. Keep the downloaded file compressed.

Step 6: Open Install from ZIP

In QGIS, open **Plugins** > **Manage and Install Plugins**, then select **Install from ZIP** in the left panel.

Step 7: Select and install the ZIP

Use the browse button to select the packaged plugin ZIP downloaded from the release assets. Click **Install Plugin** and accept the QGIS confirmation only when the package was downloaded from the official project release.

Step 8: Confirm that the plugin is enabled

After the success message appears, open the **Installed** tab and verify that the plugin is enabled. If another version was previously installed, restart QGIS before testing the updated plugin.

7.3 GitHub Repository and Release History

The public source code, documentation, issue tracker, tags, and releases are maintained at `https://github.com/ladelgadobe/BestFitInterpolation`. The repository is useful for reviewing the source code, reporting problems, and consulting version history. End users should install a packaged release ZIP rather than downloading the automatically generated *Source code* archives, because those archives may contain development files that are not part of the QGIS installation package.

Information

For reproducible work, record the plugin version, QGIS version, operating system, and installation method used in the project.

8 Opening the Plugin and Interface Overview

Before loading data, the QGIS project must be saved. This requirement gives the plugin a stable working location for the output folder, rasters, validation products, and reports created during the analysis.

Step 9: Locate the plugin in QGIS

Open `Plugins > Best Fit Interpolator > Best Fit Interpolator`. The plugin can also be opened by clicking its icon on the QGIS toolbar.

Step 10: Save the QGIS project

If the current QGIS project has not been saved, the plugin displays the message *Project must be saved*. Click `OK`, select `Project > Save As`, choose a suitable working folder, and save the project as a `.qgz` file.

Information

The output folder is created next to the saved QGIS project file. Use a descriptive project name and a working directory where the user has permission to create and modify files.

Step 11: Open the plugin again

After saving the project, open Best Fit Interpolator again from the Plugins menu or toolbar. The main window opens on the `Data` tab with a fresh session. Closing and reopening resets selections, diagnostic decisions, validation results, previews, and the Framework result. Export the outputs and record sample decisions before closing if they must be retained. The `About` tab provides the installed version, authors, manual, reference article, GitHub repository, support links, and the authors' LinkedIn profiles.



Figure 4: Data tab after loading the core inputs, with the common input controls and the Outlier diagnostic entry point.

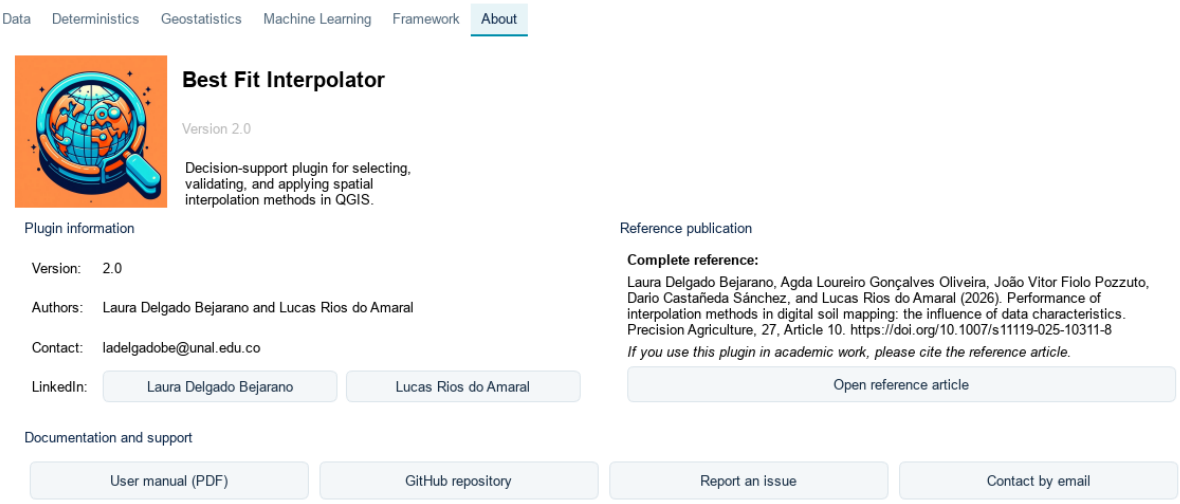


Figure 5: About page with the plugin identity, installed version, manual, article, and support links.

8.1 Interface Language

Use the `Language` selector in the upper-right corner of the plugin window to choose `English`, `Español`, or `Português`. Portuguese uses Brazilian terminology. On the first opening, the plugin follows a supported QGIS language and otherwise uses English. A manual choice is saved for later plugin sessions; it does not change the language of QGIS itself.

The interface, help dialogs, plot labels, Framework diagram, and newly generated reports use the selected language. Translations are included with the plugin and work offline. A language change requested during a calculation is applied when the active calculation finishes.

Changing language preserves loaded inputs, selected methods, parameters, validation values, and generated rasters. Layer names, field names, file paths, numerical values, and scientific symbols retain their meaning. Previously exported reports keep their original language; regenerate a report to obtain it in another language. Bibliographic references and messages supplied by external components may retain their original language. The instructions in this manual use the English control names.

9 Loading Core Data

The `Data` tab defines the common configuration shared by both independent workflows: manual interpolation and framework guidance. Before starting, load the point and polygon layers into the QGIS project. Figure 4 shows the common inputs to review before analysis.

9.1 Common Data Controls

1. **Load data points.** Select the point layer containing the sampled observations. The list is populated from point layers already loaded in the current QGIS project. Each point must have valid geometry and a usable value in the target attribute.

2. **Polygon.** Select the polygon layer that defines the study area, interpolation extent, and output mask. The polygon must be valid, loaded in QGIS, and expressed in the same projected metric CRS as the point layer.

3. **Variable and Load.** Select the numerical attribute to interpolate. The variable list is updated from the fields of the selected point layer. After selecting the point layer, polygon, and variable, click `Load`. The plugin filters incomplete or non-numerical values, calculates the spatial structure, displays the point-layer CRS, and creates the preview.

4. **Pixel size.** Enter the output raster cell size in the units of the point-layer CRS. The value entered defines the dimensions of each output pixel. For example, when the CRS uses meters, entering `5` creates pixels measuring 5 m by 5 m. Smaller values produce a finer grid but increase processing time, memory use, and output file size.

5. **Export Rasters to project folder.** Enable this option to save generated rasters in the `BestFitInterpolation` folder created beside the saved QGIS project. When it is disabled, raster outputs are temporary and should be exported manually before closing QGIS if they must be preserved.

9.2 Spatial Structure and Moran's Index

After `Load` is clicked, the `Spatial structure` line reports Moran's Index, its p-value, and one of three spatial-pattern classes:

Clustered: nearby observations tend to have similar values, providing evidence of positive spatial autocorrelation.

Dispersed: neighboring observations tend to be more dissimilar than expected under spatial randomness, providing evidence of negative spatial autocorrelation.

Random: the observed spatial arrangement does not provide sufficient evidence of either clustering or dispersion.

The plugin calculates global Moran's I using a K-nearest-neighbor spatial structure with eight neighbors, row-standardized binary weights, and a permutation-based reference distribution. The displayed p-value indicates whether the observed pattern is statistically different from spatial randomness. The final class combines the direction of Moran's I with this statistical significance: *Clustered* indicates significant positive spatial autocorrelation, *Dispersed* indicates significant negative spatial autocorrelation, and *Random* is used when the evidence is not strong enough to support either pattern.

For this reason, users should not interpret the sign of Moran's I alone. A positive or negative value only defines the direction of the pattern; the p-value determines whether that pattern is significant enough to guide the interpretation.

9.3 Outlier Diagnostic

Click `Outlier diagnostic` in `Data` to review unusually large or small values and local spatial patterns before interpolation. The window combines statistical flags, Global Moran's I, Local Moran / LISA, a point map, histogram, boxplot, and a table. Click its information icon to open a readable help window that stays open while reviewing the definitions.

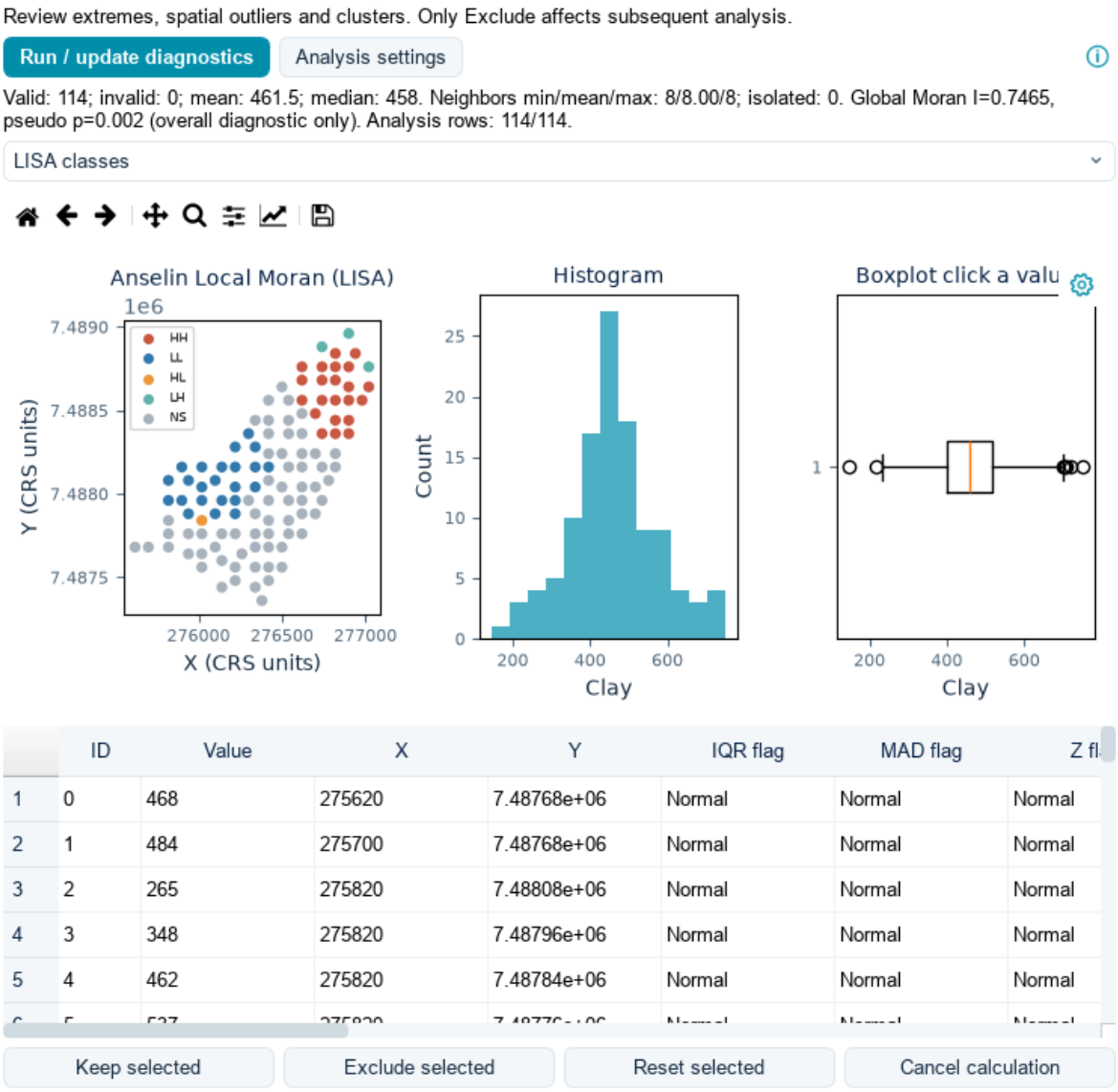


Figure 6: Outlier diagnostic: linked map, distribution plots, feature table, and explicit sample decisions.

9.3.1 Statistical Flags

Statistical flags assess a value relative to the overall distribution. IQR and MAD are selected by default; classical Z is optional and is less robust to extreme observations.

- **IQR.** With $\text{IQR} = Q_3 - Q_1$, values below $Q_1 - k\,\text{IQR}$ or above $Q_3 + k\,\text{IQR}$ are flagged. The default Tukey multiplier is $k = 1.5$.
- **MAD.** The modified score is $0.67449(x - \text{median})/\text{MAD}$, where $\text{MAD} = \text{median}(|x - \text{median}|)$. The default threshold is $|\text{score}| > 3.5$. If MAD is zero, values equal to the median score zero; unequal values receive signed infinity and require review.
- **Z.** The classical score uses the mean and standard deviation. Its default threshold is $|Z| > 3$.
- **Statistical consensus.** `Any` flags a point if any selected method flags it. `All` requires all selected methods. `At least N` requires the chosen minimum number of flags. These settings combine statistical methods; spatial evidence is reported separately.

9.3.2 Global and Local Spatial Evidence

Global Moran's I describes the overall spatial pattern. It does not identify an individual point as an outlier. Local Moran / LISA evaluates each point relative to its neighbors, following the local association approach of ANSELIN (1995). With centered values $z_i = x_i - \bar{x}$, the local statistic is

$$I_i = \frac{z_i}{m_2} \sum_j w_{ij} z_j, \qquad m_2 = \frac{1}{n} \sum_i z_i^2.$$

The plugin uses row-standardized binary neighborhood weights and conditional permutations that keep the focal value fixed while sampling possible neighbors from the other observations.

Table 5: Reading Local Moran / LISA classes. H and L mean above and below the analysis mean.

Class	**Point and neighborhood**	**Interpretation**
HH	High point, high neighbors	Significant cluster of high values
LL	Low point, low neighbors	Significant cluster of low values
HL	High point, low neighbors	Potential spatial outlier
LH	Low point, high neighbors	Potential spatial outlier
NS	No significant local pattern	Insufficient local evidence at the chosen alpha; statistical flags may still apply
Invalid	Unusable value or geometry	Cannot contribute to the analysis

The first letter describes the point and the second the mean of its neighbors. HH and LL are clusters, whereas HL and LH are the spatial-outlier classes. A significant local contrast may represent a genuine boundary, management effect, or environmental transition; it is not automatically a measurement error.

9.3.3 Analysis Settings

Click `Analysis settings` to open the parameter window. The statistical controls identify unusually large or small values; the neighborhood and LISA controls assess local spatial patterns. Table 6 explains every control shown in Figure 7.

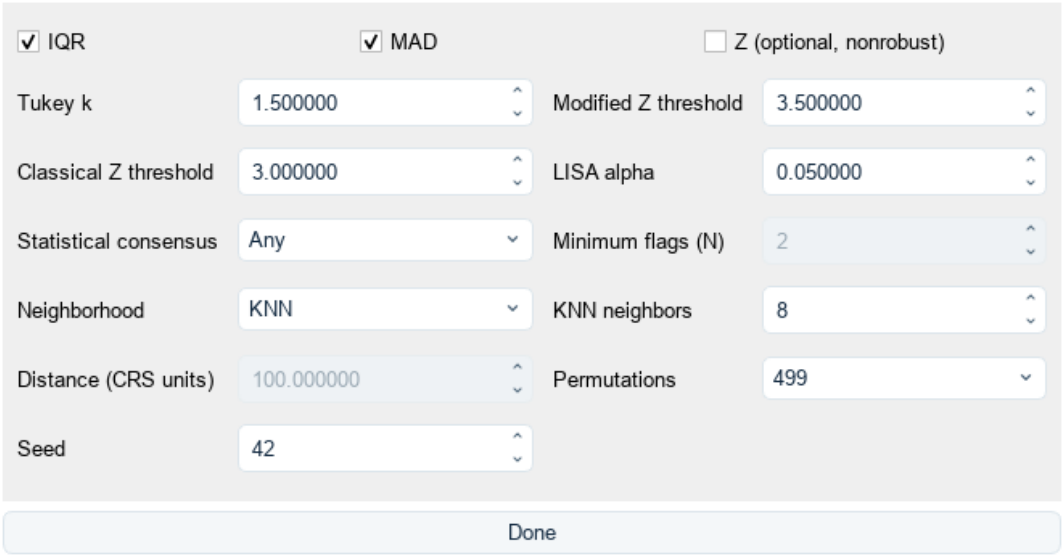


Figure 7: Analysis settings for statistical thresholds, consensus, neighborhood, permutations, alpha, and seed.

Table 6: Outlier diagnostic parameters and their interpretation.

Control	Default	Meaning and use
`IQR`, `MAD`, `Z`	IQR and MAD	Select which statistical methods contribute to the consensus. Z is optional and nonrobust. These checkboxes do not turn LISA on or off.
`Tukey k`	1.5	Multiplier of the interquartile range in the IQR limits $Q_1 - k\,\mathrm{IQR}$ and $Q_3 + k\,\mathrm{IQR}$. Larger k widens the limits and flags fewer values; smaller k flags more.
`Modified Z threshold`	3.5	Absolute-score cutoff for MAD, using the median and median absolute deviation. A value is flagged when its modified score exceeds this cutoff in magnitude. A higher cutoff flags fewer values.
`Classical Z threshold`	3.0	Absolute-score cutoff for the optional Z method: $Z = (x - \text{mean})/\text{standard deviation}$. Unlike MAD, it uses the mean and standard deviation, which extreme values can distort. A higher cutoff flags fewer values.
`LISA alpha`	0.05	Maximum local pseudo p-value accepted as significant. Lower alpha requires stronger evidence; higher alpha accepts weaker evidence. It controls LISA classes, independently of the statistical thresholds.
`Statistical consensus`	Any	`Any`: at least one selected method must flag the point. `All`: every selected method must flag it. `At least N`: require a specified number of selected methods.
`Minimum flags (N)`	2	Active only with `At least N`. With IQR and MAD selected, $N = 2$ requires both flags. Choose N no larger than the number of selected methods; otherwise no point can meet the consensus.
`Neighborhood`	KNN	Choose `KNN` for a fixed neighbor count, or `Distance threshold` for a fixed radius. This changes the spatial context used by Global Moran and LISA.

Control	Default	Meaning and use
`KNN neighbors`	8	Number of nearest other points used in KNN mode, limited by the available observations. Smaller counts emphasize local structure; larger counts represent a broader neighborhood.
`Distance (CRS units)`	100	Radius used only in `Distance threshold` mode. In a metric CRS, 100 means 100 m. A small radius can leave points isolated; check the isolated count before interpreting LISA.
`Permutations`	499	Random rearrangements used for the significance test; choices are 99, 499, and 999. More permutations improve pseudo p-value resolution and take longer. The minimum possible pseudo p-value is $1/(P+1)$.
`Seed`	42	Initializes the random generator. Keep the same inputs, neighborhood, permutations, and seed to reproduce a diagnostic run. Changing the seed can change the estimated pseudo p-values.
`Done`		Closes the settings window. Return to the diagnostic window and click `Run / update diagnostics` to calculate results using the new settings.

Start with the defaults, inspect the flags and neighboring values, and then adjust a parameter only when its interpretation fits the sampling design. Changing a threshold changes a diagnostic label; it does not establish that a measurement is wrong or automatically exclude it.

Information

Local pseudo p-values are not adjusted for multiple comparisons. Use the map and table as an exploratory diagnostic and interpret the evidence with sampling design and field knowledge. The configurable Global Moran result in this window may differ from the automatic `Data` summary because its neighborhood, seed, and permutation settings can differ.

9.3.4 Reviewing and Applying Sample Decisions

1. Load the common inputs in `Data` and open `Outlier diagnostic`.
2. Review `Analysis settings`, then click `Run / update diagnostics`. Inspect the valid count, missing values, neighbor counts, and isolated points.
3. Switch between `LISA classes` and `Target values`. Select rows or points to link the table, map, and QGIS feature selection; the distribution plots provide another way to inspect extreme values.
4. Check the original record, field notes, units, and neighboring measurements before deciding. For example, an HL point with an IQR flag deserves inspection, while an HH cluster may describe a real high-value zone.
5. Use `Keep selected` to record retention, `Exclude selected` to omit the selected usable points from subsequent analysis, or `Reset selected` to clear the decision. Only Exclude changes the analysis mask; flags alone do not exclude points.
6. Recalculate the diagnostics and rerun fitting, validation, interpolation, and comparisons as needed after changing the analysis population. Review the updated sample count before accepting results.

The original point layer is preserved. Keep cannot make invalid values or missing geometries usable. Decisions belong to the selected layer and target variable within the current plugin session;

record the feature IDs and reasons before closing. Changing the data invalidates diagnostic and model snapshots so that an earlier result is not silently reused.

9.4 Coordinate Reference System

The `CRS` line displays the coordinate reference system of the selected point layer. The CRS is not configured or transformed inside the plugin. Input layers must be prepared in QGIS before loading them.

Information

Use a projected CRS whose map units are meters, such as the appropriate local UTM zone. Do not use a geographic latitude–longitude CRS such as EPSG:4326. Geographic coordinates are expressed in degrees, so distance calculations, Moran's Index, pixel size, and distance-based interpolation can fail or produce incorrect results. Reproject the point and polygon layers to the same metric CRS before continuing.

If the selected point layer does not use meter units, the plugin displays the warning *Metric CRS required*. The plugin also checks whether the point and polygon layers use compatible CRSs and whether valid sample points overlap the selected polygon. A warning is shown in the `Data` tab, and interpolation is blocked if the spatial inputs are incompatible or if no usable samples fall inside the polygon. Resolve these problems before running interpolation or validation.

9.5 Preview

The preview displays the sample points colored according to the selected variable together with the boundary polygon. Use it to verify that the correct layers and variable were selected, that the samples fall within the intended study area, and that the spatial distribution and value range appear reasonable.

10 Deterministic Interpolation: IDW and TPS

After the common inputs have been loaded in the `Data` tab, open `Deterministics`. This tab provides three mutually exclusive interpolation choices: Inverse Distance Weighting (IDW) with manual parameters, IDW with automatic optimization, and Thin Plate Spline (TPS). At least ten valid observations are required to run an interpolation.

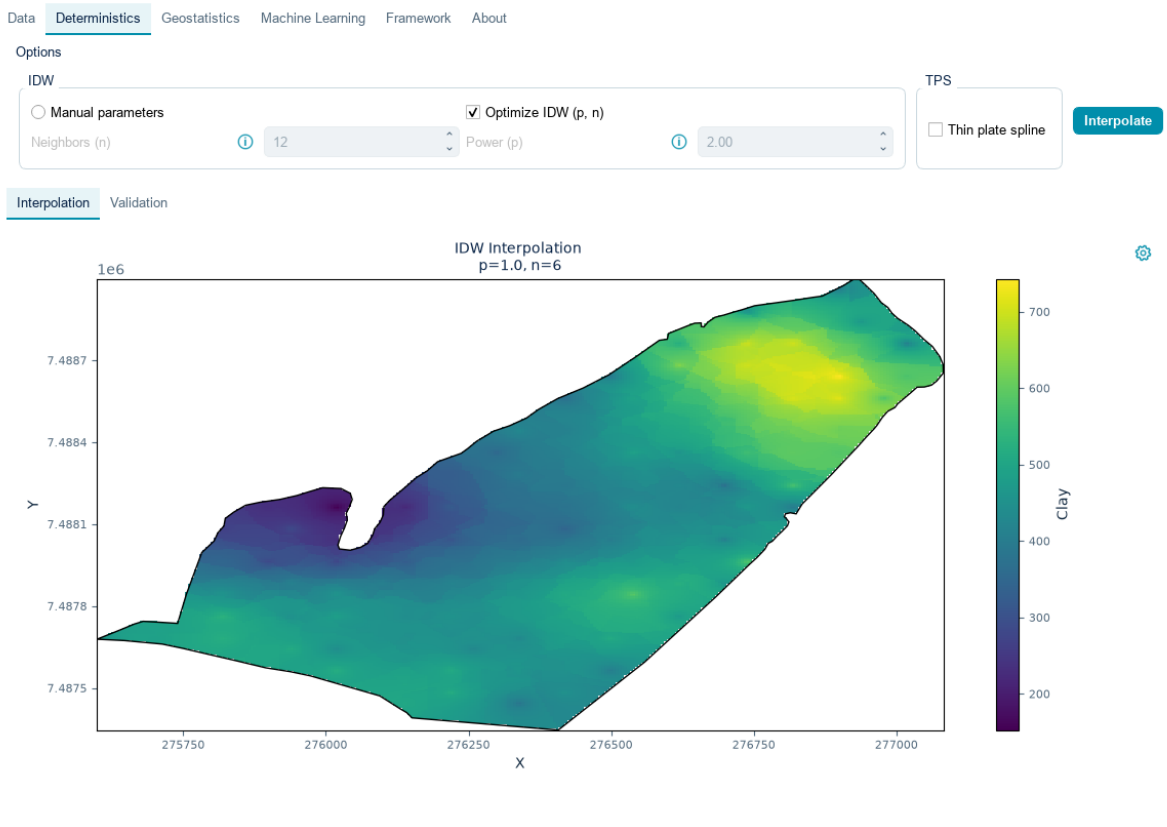


Figure 8: Deterministics tab showing the available interpolation methods and the result window for the selected method.

10.1 Available Interpolation Options

1. **Manual IDW.** Select `Manual parameters`, then enter the neighbor count n and power p. The neighbor count controls the observations used for each prediction. Smaller neighborhoods emphasize local variation; larger neighborhoods generally produce smoother surfaces. Increasing p gives nearby observations more influence. Decreasing p spreads influence across the neighborhood.

2. **Optimized IDW.** Select `Optimize IDW (p, n)` when the plugin should choose both parameters automatically. The manual p and n controls are disabled in this mode. The plugin evaluates candidate p and n combinations using five-fold cross-validation and selects the combination with the smallest Interpolation Selection Index (ISI), which combines normalized mean absolute error and the variability of prediction errors (BIER; SOUZA, 2017).

3. **Thin Plate Spline (TPS).** Select `Thin plate spline` to create a smooth surface using a radial-basis thin-plate spline. TPS does not expose additional parameters in this tab. For normal datasets it uses the global thin-plate spline. Dense and massive profiles use a local neighborhood, as described in Section 19. It requires unique sampling coordinates; duplicate point coordinates must be resolved before interpolation.

4. **Run the selected method.** Click `Interpolate`. The plugin reuses the point layer, target variable, polygon, pixel size, and export option defined in the `Data` tab. The resulting surface is shown in the `Interpolation` subtab, clipped to the selected polygon, and added to QGIS. When raster export is enabled, the file is also written to the project output folder.

Information

Manual IDW, optimized IDW, and TPS cannot run simultaneously. Selecting one option automatically deactivates the others. The displayed interpolation map is a visual result; method performance must be assessed in the `Validation` subtab using the same configuration that produced the interpolation.

11 Geostatistical Interpolation: Ordinary and Universal Kriging

After loading the common inputs, open the `Geostatistics` tab. The `Type` selector offers `Ordinary Kriging` and `Universal Kriging (linear)`. Both use the variable, polygon, pixel size, and export configuration loaded in `Data`. The fitting method, MoM or REML, controls estimation of the spatial covariance parameters; it is separate from the choice between ordinary and universal kriging.

11.1 Choosing Ordinary or Universal Kriging

Ordinary Kriging assumes an unknown constant mean over the prediction neighborhood. Universal Kriging represents a linear spatial mean,

$$m(x, y) = \beta_0 + \beta_x x + \beta_y y,$$

and models the remaining spatial dependence. In this implementation, the drift uses the coordinates `x` and `y`; it does not use the environmental raster predictors of the RF-based Regression Kriging workflow. The sample locations must support estimation of this plane: collinear locations cannot identify a two-dimensional linear trend.

Select the kriging type before calculating or fitting the variogram. With Universal Kriging, the MoM experimental semivariogram describes residuals after removing the fitted linear trend; REML accounts for the linear trend in covariance estimation. MoM shows experimental points and the fitted curve, while REML shows the theoretical curve. Review the parameters and click `Interpolate`. Changing the type clears the dependent geostatistical results, so refit and rerun validation for the new choice.

The standalone Geostatistics validation supports the selected Universal Kriging model and refits its trend and covariance within each training partition. Inspect its predictive metrics before choosing it over Ordinary Kriging. The Framework decision tree retains its original `OK` alternative, which means Ordinary Kriging; choosing Universal Kriging in the standalone tab does not change the Framework method.

Prediction, conditional standard deviation, and variance outputs are described in Section 13.

11.2 Automatic Choice Between MoM and REML

When `Fit method` is set to `Automatic`, the plugin uses REML for fewer than 100 valid samples when the REML backend is available, and MoM for 100 samples or more. This implementation rule is consistent with the sampling evidence discussed by KERRY; OLIVER (2007): empirical semivariograms calculated by MoM commonly require approximately 100 appropriately spaced observations, whereas REML can provide useful parameter estimates with smaller samples. The threshold is a practical plugin rule, not a universal statistical boundary; sampling design, spatial structure, and data quality still matter.

The user can override the automatic choice by selecting `MoM` or `REML`. Manual REML is available for datasets with fewer than 500 valid samples. For 500 samples or more, the plugin uses MoM because REML optimization can become computationally expensive.

11.3 Method of Moments (MoM)

Figure 9 shows the MoM workspace: experimental semivariogram, fitted theoretical model, spatial-dependence summary, and resulting Ordinary Kriging map.

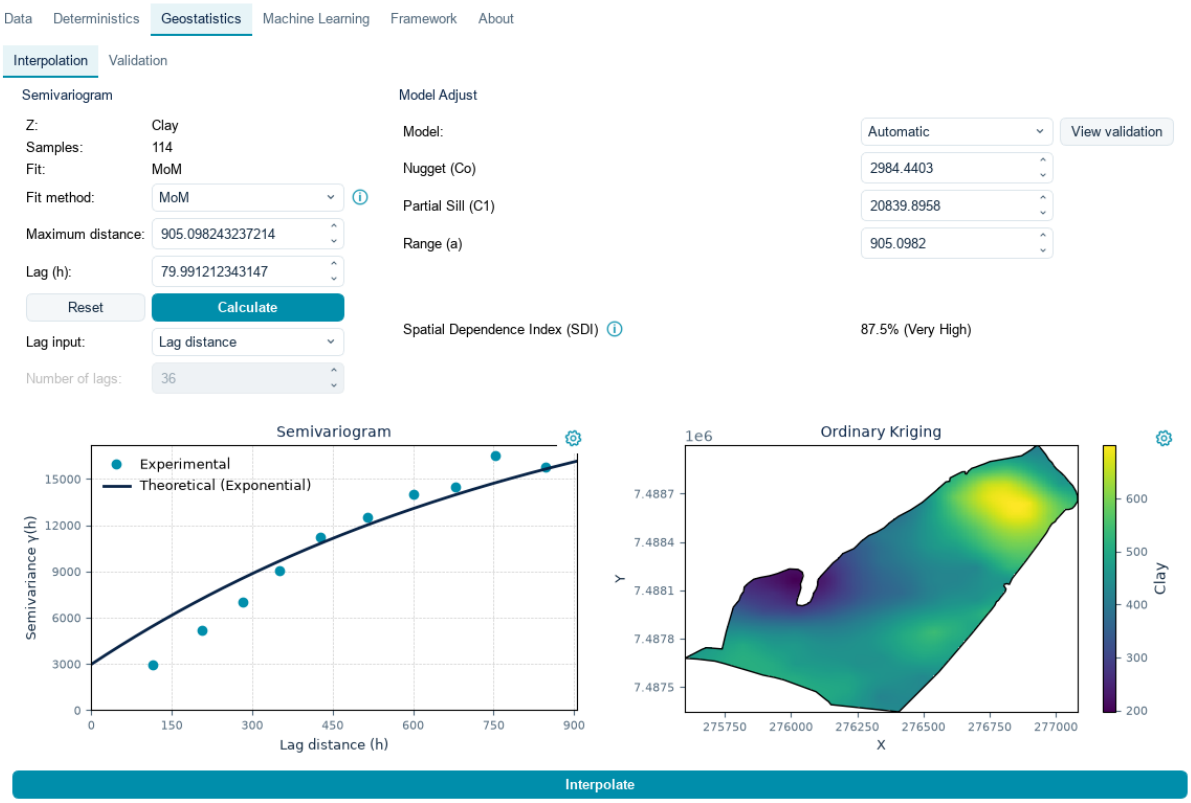


Figure 9: Geostatistics tab using MoM with an experimental semivariogram, a fitted theoretical model, the Spatial Dependence Index, and the Ordinary Kriging map.

The `Semivariogram` panel reports the target variable (`Z`), number of valid samples, and selected fitting method. `Maximum distance` defines the greatest point-pair separation used to construct the experimental semivariogram, while `Lag (h)` defines the distance-class width. Click `Calculate` after changing either value. `Reset` restores the automatically proposed configuration.

Under `Model Adjust`, the plugin can evaluate three theoretical semivariogram models:

- **Spherical:** increases with distance and reaches its sill at the specified range.
- **Exponential:** approaches the sill progressively and can represent spatial correlation that decays gradually.
- **Gaussian:** is smooth near the origin and is suitable when short-distance variation changes gradually.

With `Model` set to `Automatic`, the plugin evaluates the Spherical, Exponential, and Gaussian models by leave-one-out cross-validation. In Geostatistics, it ranks the candidates by the highest Lin's concordance correlation coefficient (LCCC), then the lowest root mean square error (RMSE), and then the highest R^2 if a further tie remains. Click `View validation` to inspect RMSE, RMSE%, MAE, R^2, Pearson correlation, and Lin's concordance correlation coefficient for the three candidate models. Select a successful model row when another candidate is preferable. The Framework semivariogram popup uses its own R^2/RMSE ranking, described later in this manual.

The automatic MoM procedure jointly fits the nugget, partial sill, and model scale by constrained weighted least squares. For each candidate scale, nugget and partial sill are estimated together, and a bounded iterative search refines the scale. The distance-class weights are

$$w_j = \frac{\text{median}(h)}{\text{median}(h) + h_j},$$

where h_j is the representative distance of lag class j, and the median uses the lag-class distances included in the fit. Shorter distances receive greater weight. The nugget is nonnegative; the partial sill and scale are positive. These are variogram-fitting weights, separate from the spatial prediction weights used by kriging.

The model can also be selected manually. Adjust `Nugget (C0)`, `Partial Sill (C1)`, and `Range (a)` only when the experimental semivariogram and subject-matter knowledge support the values. The nugget represents unresolved variation at short positive separation, and the partial sill is the spatially structured variance. The exact-observation convention and its uncertainty interpretation are described in Section 13.1.

The blue points in the left plot are the experimental semivariances calculated from sample pairs, and the black curve is the fitted theoretical model. The right plot previews the Ordinary Kriging surface inside the selected polygon. After reviewing the model, click `Interpolate` to create the final raster.

11.4 Model Scale and Practical Range

For both MoM and REML, `Range (a)` is the raw model-scale parameter in the distance units of the layer CRS. It is not the same numerical distance for every covariance model. The interface and reports also identify the conventional practical range:

Table 7: Relation between the model scale a and the conventional practical range.

Model	Practical range	Interpretation
Spherical	a	The full sill is reached at this distance.
Exponential	$3a$	Approximately 95% of the structured semivariance is reached.
Gaussian	$\sqrt{3}\,a$	Approximately 95% of the structured semivariance is reached.

For example, an exponential model with `Range (a)` equal to 100 m has a conventional practical range of approximately 300 m. Enter the model scale in the parameter box; do not enter 300 m when the intended exponential scale is 100 m. The exponential and Gaussian curves approach the sill asymptotically, so their practical ranges are conventional distances rather than exact cutoffs. This convention also applies to the Framework semivariogram and the residual variogram in RK.

11.5 Spatial Dependence Index

The plugin calculates the Spatial Dependence Index (SDI) from the fitted nugget and partial sill:

$$\mathrm{SDI} = \frac{C1}{C0 + C1} \times 100\%.$$

Table 8: Spatial Dependence Index classification used by the plugin.

SDI (%)	Classification
$0 \leq \mathrm{SDI} < 20$	Very Low
$20 \leq \mathrm{SDI} < 40$	Low
$40 \leq \mathrm{SDI} < 60$	Moderate
$60 \leq \mathrm{SDI} < 80$	High
$80 \leq \mathrm{SDI} \leq 100$	Very High

A low SDI indicates that the nugget accounts for a large share of the total fitted variance. A high SDI indicates that the spatially structured component is dominant. SDI describes the fitted

spatial-dependence structure; it is not, by itself, a measure of predictive accuracy. In Figure 9, the displayed SDI describes this teaching dataset and should be read from the current fitted parameters.

11.6 Restricted Maximum Likelihood (REML)

Restricted maximum likelihood (REML) estimates covariance parameters directly from the observations by maximizing a restricted likelihood. It does not require fitting experimental semivariogram points. Consequently, the REML view displays the fitted theoretical curve and parameter estimates, without experimental points, for the selected kriging type.

REML can be advantageous for smaller datasets because it accounts for parameter estimation through a likelihood-based procedure, but its numerical optimization is more demanding than MoM. For this reason, the plugin recommends REML automatically only below 100 valid samples when its backend is available, permits manual REML below 500 samples, and avoids it for larger datasets.

Information

The `View validation` button compares theoretical semivariogram models using the current fitting method. REML shows only a theoretical curve; changing between MoM and REML refreshes the corresponding view. It is different from the common `Validation` subtab used later to evaluate the predictive performance of the completed interpolation.

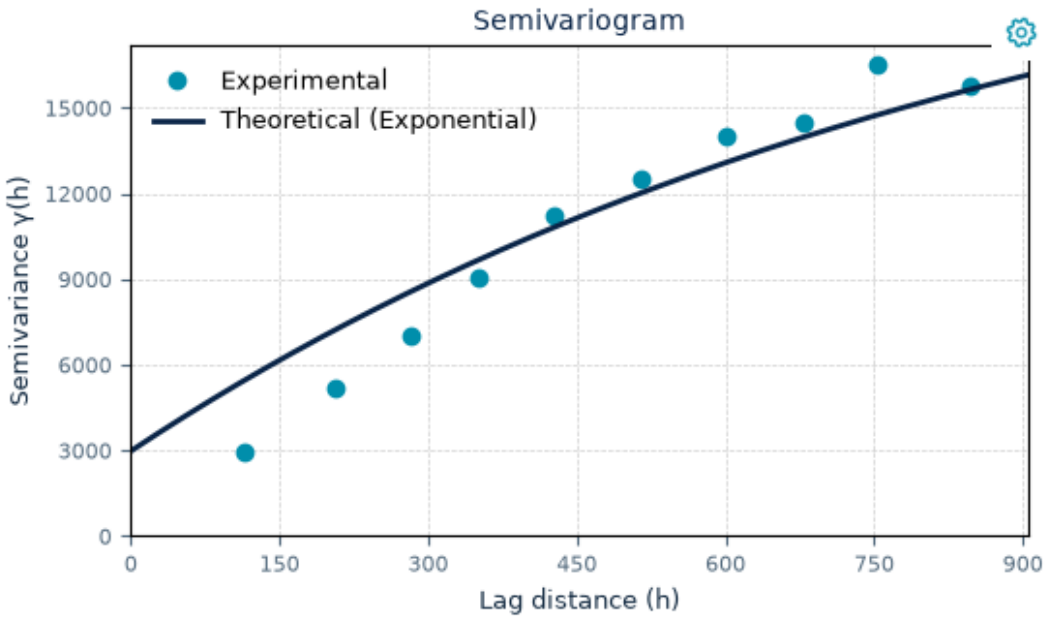


Figure 10: MoM view: experimental semivariances and fitted theoretical curve.

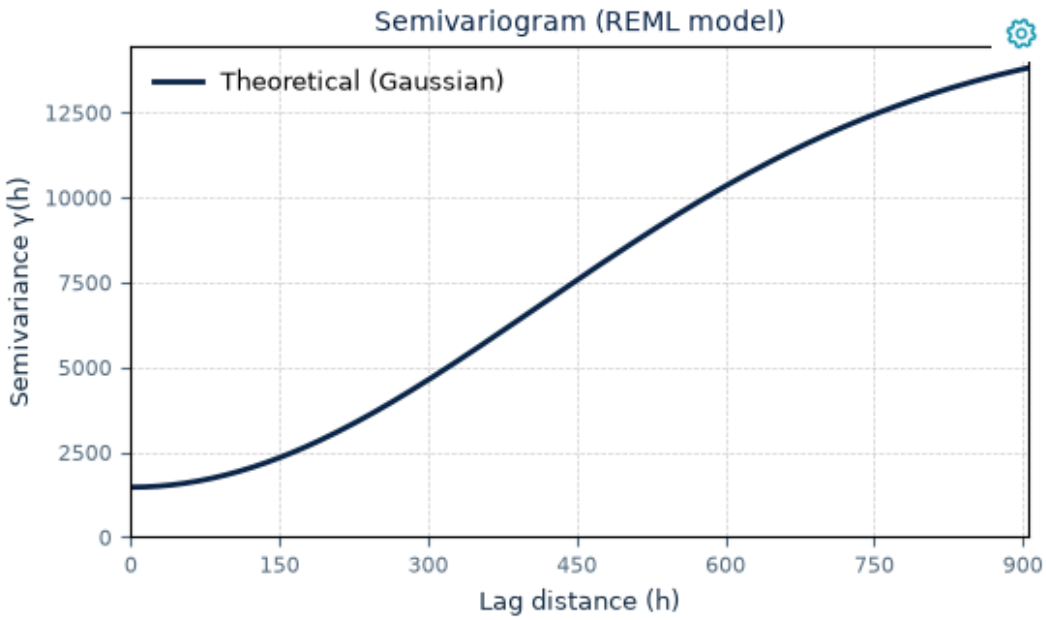


Figure 11: REML view: theoretical model only; no experimental semivariogram is displayed.

12 Machine Learning Interpolation

After the common inputs have been loaded in the `Data` tab, open the `Machine Learning` tab when the interpolation should use environmental predictors in addition to the sampled target variable. In this workflow, the point observations provide the response values, while raster covariates provide predictor variables that help explain spatial variation across the study area. These covariate rasters must first be loaded in the QGIS project environment; the plugin reads the raster layers available in QGIS and does not create or download covariates automatically.

12.1 Covariates

Covariates are raster layers that describe auxiliary spatial information related to the target variable. Examples may include elevation, terrain derivatives, spectral indices, soil maps, climate layers, or any other continuous raster predictor prepared by the user. Because the plugin works inside QGIS, the covariate rasters should be added to the QGIS layer panel before they are selected in the `Machine Learning` tab.

Figure 12 shows the covariates window used before fitting machine learning models. The same tab brings together raster loading, covariate preprocessing, resampling, standardization, extraction of raster values to sample points, and correlation diagnostics.

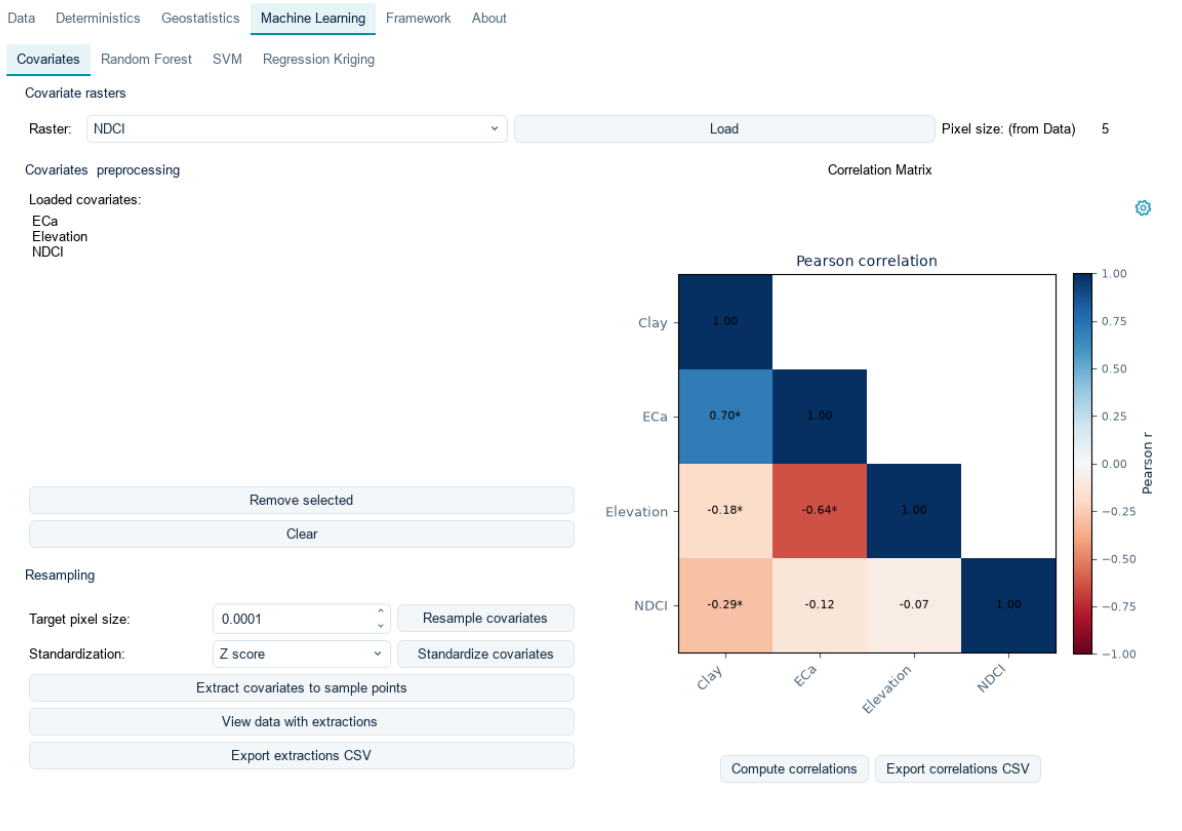


Figure 12: Machine Learning covariates tab showing raster loading, covariate preprocessing, resampling, standardization, extraction controls, and the Pearson correlation matrix.

1. **Load covariate rasters.** Select each raster layer from the QGIS project and click `Load`. The loaded-covariates list is updated as rasters are added, removed, cleared, resampled, or standardized.

2. **Check spatial compatibility.** Covariates should use the same projected CRS as the point and polygon layers. They should also have compatible extent, resolution, and alignment so that the value extracted at each sample point represents the same spatial support.

3. **Resample covariates when needed.** Use `Resample covariates` when the raster predictors have different pixel sizes, extents, or grid alignment. The target pixel size can follow the output pixel size defined in the `Data` tab or another value selected by the user in the covariates window. Resampling creates a consistent predictor stack, so each grid cell used for interpolation is described by covariate values with the same spatial support.

4. **Standardize covariates when appropriate.** The `Standardization` control provides two options: `Z-score` and `Range [-1,1]`. `Z-score` centers each covariate and expresses values relative to its variability, while `Range [-1,1]` rescales values to a common interval from -1 to 1. These transformations help place covariates with different units and numerical ranges on comparable scales, which is especially useful for scale-sensitive models such as Support Vector Machine.

5. **Extract covariates to sample points.** After the covariates are prepared, extract their raster values at the sampled point locations. These extracted values form the predictor table used to train the machine learning models.

6. **Review covariate correlations.** Use the correlation matrix as a diagnostic tool to inspect relationships among the target variable and the selected covariates. Strong correlations between covariates may indicate redundant predictors, while weak relationships with the target variable may reduce model usefulness.

Information

Machine learning interpolation depends strongly on covariate quality. Covariates with missing values, mismatched CRS, incomplete coverage, weak relationship with the target variable, or excessive redundancy can reduce model performance even when the algorithm runs successfully.

12.2 Available Machine Learning Methods

The plugin includes Random Forest, Support Vector Machine (SVM), and Regression Kriging workflows. The following model-specific items describe the parameters available in the interface, the validation controls, and the expected outputs.

Information

The first time the plugin is installed, or the first time a machine learning method such as RF or SVM is executed in a QGIS environment without the required Python packages, the plugin may request permission to install the missing machine learning dependencies automatically. This step prepares the QGIS Python environment for methods that use `scikit-learn`. After the dependencies are available, the same installation prompt should not appear again for that environment.

12.2.1 Random Forest

In the plugin, Random Forest (RF) is implemented as a spatial machine learning interpolation method. The model can use the point coordinates, represented by `x` and `y`, together with the raster covariates loaded in the `Covariates` tab. In this way, RF can learn from both the spatial position of the samples and the environmental predictors prepared in QGIS.

Before running RF, load and review the predictor rasters in `Covariates`, including their coverage, alignment, and any resampling or standardization. The extraction tool lets the user inspect predictor values at sample locations; the interpolation workflow also reads these values when building its training table. When a standalone RF interpolation or hyperparameter search starts, the predictor-selection dialog allows the user to include or omit `x`, `y`, and the loaded raster covariates. Only the selected predictors are used for training, search, and map prediction.

Figure 13 shows the RF interpolation window. The upper panel contains the manual and Grid Search controls, while the lower panels show the variable-importance plot and the RF interpolation map produced after the model is fitted.

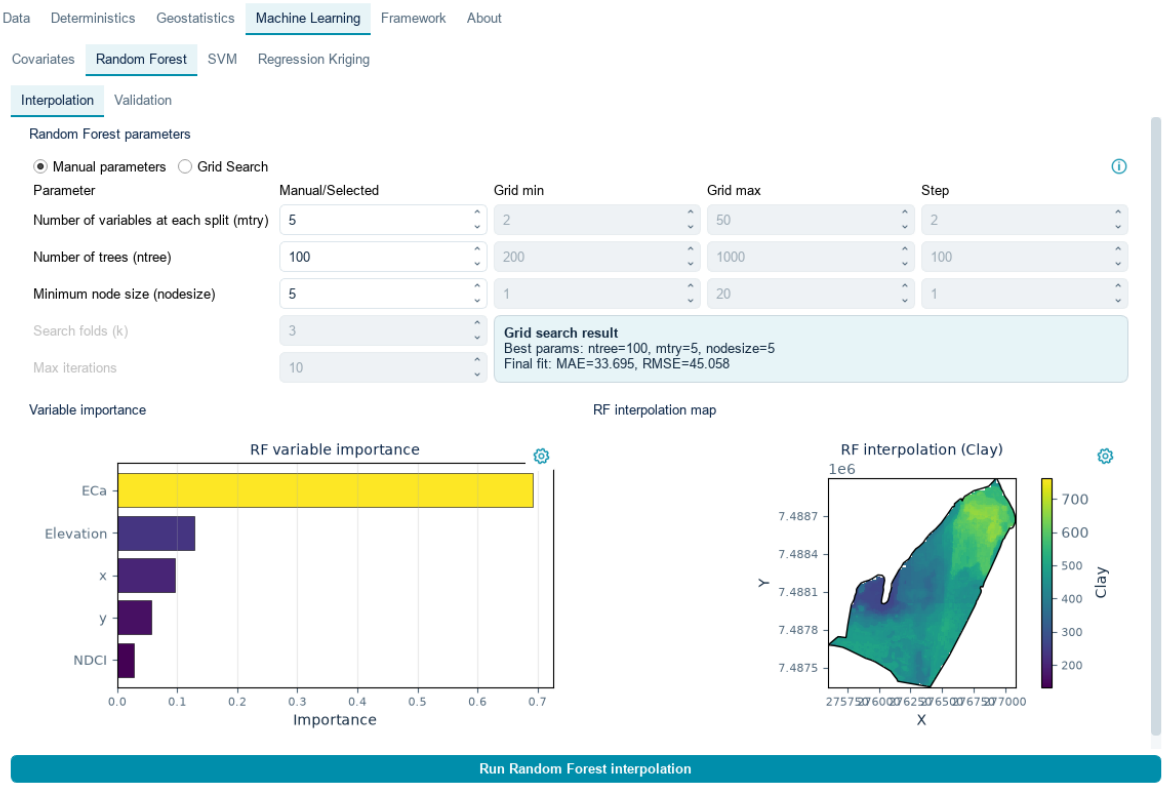


Figure 13: Random Forest interpolation tab showing parameter controls, Grid Search settings, variable importance, and the interpolated RF map.

The `Random Forest` tab contains two parameter modes, shown in the upper panel of Figure 13:

1. **Manual parameters.** The user directly defines `mtry`, `ntree`, and `nodesize`. In this mode, the plugin fits the RF model once using the selected predictors and the parameter values entered by the user.

2. **Grid Search.** The user defines a search range for each RF parameter using `Grid min`, `Grid max`, and `Step`. The plugin builds candidate values from those ranges and evaluates parameter combinations until the `Max iterations` limit is reached. The `Search folds (k)` value controls how many cross-validation folds are used internally to compare candidate parameter sets during the search. Grid Search is therefore used to find suitable RF hyperparameters for the selected predictor set; it does not select covariates by itself.

The RF parameters in Figure 13 control different parts of the model:

- `mtry`: number of predictor variables randomly considered at each split in each tree. If many predictors are selected, a smaller `mtry` makes the trees more different from each other; a larger `mtry` lets each split evaluate more variables and may favor dominant predictors.

- `ntree`: number of trees grown in the forest. Increasing `ntree` usually makes the prediction more stable because the final map averages more trees, but it also increases processing time and memory use.

- `nodesize`: minimum number of samples allowed in a terminal node. Smaller values allow trees to follow finer local patterns in the training data; larger values make each tree smoother and more generalized.

In Grid Search, the `Step` value determines how the candidate list is created between the minimum and maximum values. A small step explores the range more finely but creates more candidate

combinations, while a larger step reduces processing time by testing fewer values. The same logic applies separately to `mtry`, `ntree`, and `nodesize`.

After the model is fitted, the lower panels of Figure 13 display the RF interpolation map and the variable-importance plot. The importance plot helps identify which selected predictors contributed most to the RF model. This result should be interpreted together with the covariate correlation matrix because a variable can appear important while still being redundant with another predictor.

Information

RF results depend on predictor quality and relevance. Coordinates can represent location-related variation, but raw `x` and `y` can also produce orthogonal artifacts in RF maps. Inspect both validation performance and the resulting spatial patterns. The standalone workflow lets the user omit coordinates. Weak, redundant, poorly aligned, or incomplete covariates can reduce performance. RF tree-dispersion outputs are explained in Section 13.2.

12.2.2 Support Vector Machine

Support Vector Machine (SVM) is also implemented as a spatial machine learning interpolation method in the plugin. As with RF, the model can use the point coordinates, `x` and `y`, together with the raster covariates prepared in the `Covariates` tab. The selected predictors are used to train the model and to generate predictions over the interpolation grid.

Before running SVM, complete the covariate preparation workflow and select the final predictors when the plugin asks which variables should be used. The selected set may include only raster covariates, only coordinates, or a combination of `x`, `y`, and raster covariates, depending on the interpolation objective and the available information.

Figure 14 shows the SVM interpolation window. The left panel contains the manual and Grid Search controls, and the right panel shows the SVM interpolation map produced after fitting the model.

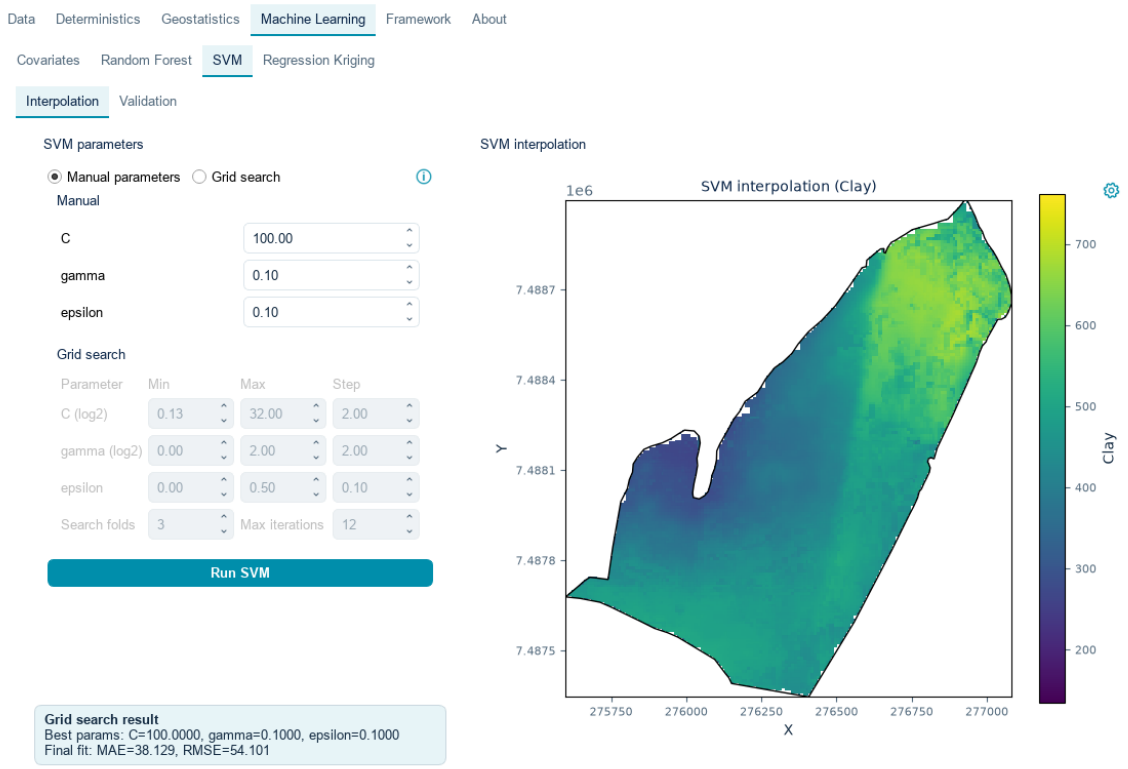


Figure 14: SVM interpolation on Paulínia with illustrative manual settings C=100, gamma=0.1, and epsilon=0.1. Validate or tune these settings for the mapping objective.

The SVM tab contains two parameter modes, shown in the left panel of Figure 14:

1. **Manual parameters.** The user directly defines C, gamma, and epsilon. In this mode, the plugin fits the SVM model once using the selected predictors and the parameter values entered by the user.
2. **Grid Search.** The user defines minimum, maximum, and step values for C, gamma, and epsilon. The plugin builds candidate parameter values and evaluates combinations until the Max iterations limit is reached. The Search folds value controls how many folds are used internally to compare candidate parameter sets during the search. Grid Search is used to tune SVM hyperparameters for the selected predictor set; it does not select covariates automatically.

The SVM parameters in Figure 14 control how flexible or smooth the fitted model will be:

- C: penalty applied to prediction errors. Higher C values force the model to fit the training data more strictly, while lower values allow a smoother model with more tolerance to errors.
- gamma: controls how local the influence of each sample is in the radial basis function kernel. Higher gamma values create more local and complex responses; lower values produce smoother and broader spatial patterns.
- epsilon: defines the insensitive margin around the regression function. Errors smaller than epsilon are ignored during training. Smaller values fit the data more tightly, while larger values produce a smoother model.

In the Grid Search table, C and gamma are defined on a log2 scale, while epsilon is defined on its original scale. The Step value controls how many candidate values are created between the minimum and maximum settings. Smaller steps explore the search space more finely but increase processing time; larger steps reduce the number of combinations tested.

Information

SVM is sensitive to predictor scale. The current fitting pipeline standardizes the selected predictors internally. The Covariates tab also provides explicit Z-score and Range [-1,1] tools for preparing shared inputs; record any transformation used. Predictor scaling does not replace parameter tuning and independent validation.

12.2.3 Regression Kriging

Regression Kriging (RK) is implemented in the plugin as a hybrid interpolation method. It combines a Random Forest trend model with residual kriging. In the first stage, RF uses the selected predictors, such as x, y, and raster covariates, to model the deterministic or explanatory component of the target variable. In the second stage, the residuals from the RF model are interpolated with kriging to account for spatial structure that remains after the covariate-based prediction.

Before running RK, complete the Covariates workflow and select the final predictors when the plugin requests them. Because the RF stage is the base of RK, the predictor-selection logic is the same as in the RF tab: only the selected variables are used to fit the trend, estimate residuals, tune RF parameters, and build the final map.

Figure 15 shows one Regression Kriging workspace with the RF trend and residual kriging panels side by side. The diagnostic area includes the residual semivariogram and variable importance, beside the RK map. The fitting controls belong to Interpolation; Validation is used to assess the configured RK result.

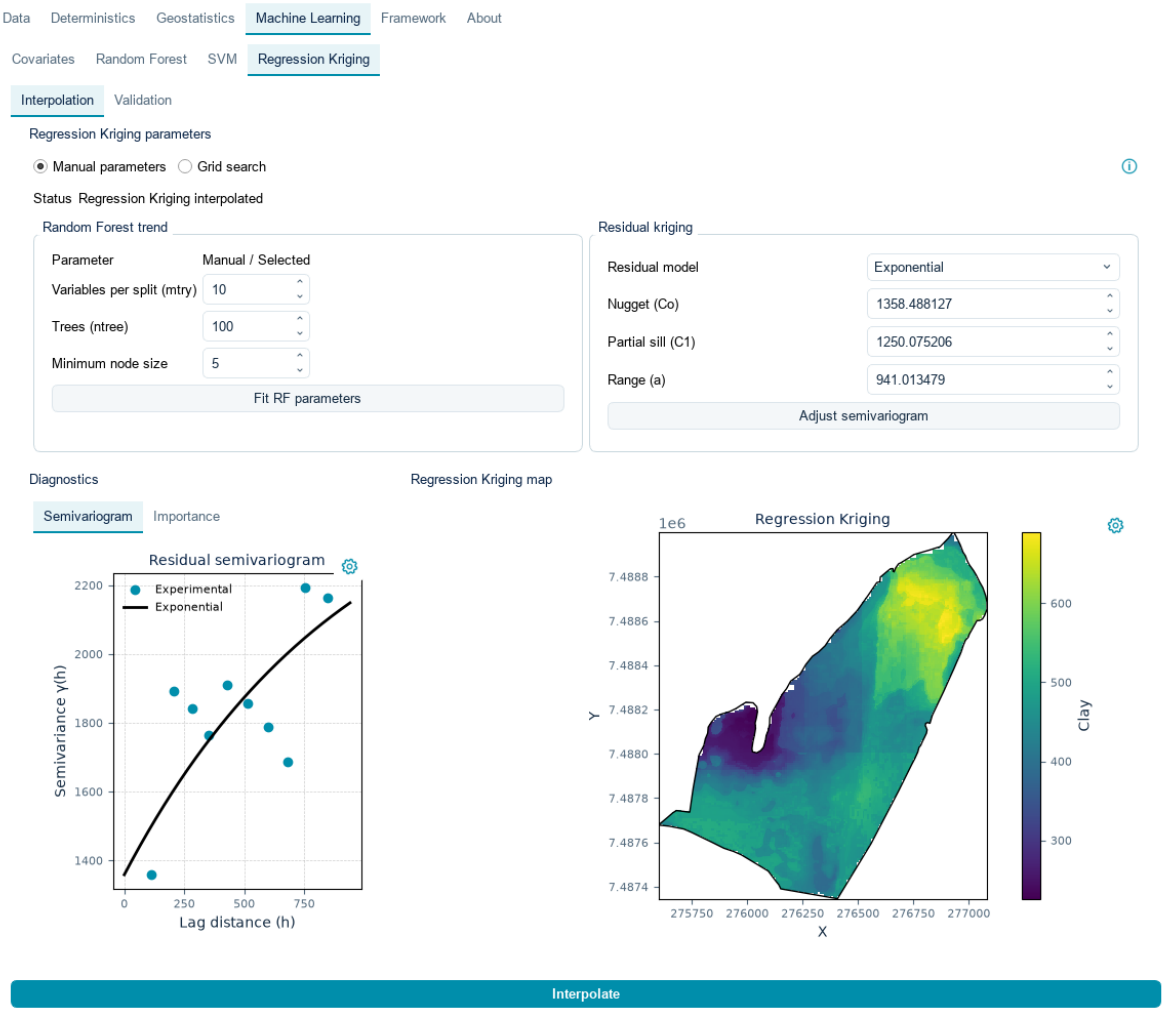


Figure 15: Regression Kriging tab showing the Random Forest trend settings, residual kriging parameters, residual semivariogram, and final RK interpolation map.

The RK workflow has two main parameter groups:

1. **Random Forest trend.** The RF component uses `mtry`, `ntree`, and `nodesize`, with the same interpretation described for the RF method. These parameters can be entered manually or searched using `Grid search`. The `Fit RF parameters` button fits or tunes the RF trend and reports the selected RF parameter combination.

2. **Residual kriging.** After the RF trend is fitted, the plugin models the residuals using a residual variogram. The user can choose the residual variogram model and review or adjust the kriging parameters before clicking `Interpolate`.

The residual kriging controls in Figure 15 describe the spatial dependence left in the RF residuals:

- `Residual variogram model`: theoretical model fitted to the residual semivariogram. The interface provides `Spherical`, `Exponential`, and `Gaussian` options.
- `Nugget (Co)`: residual semivariance at very short distances, including measurement error or micro-scale variation not captured by the RF trend.
- `Partial sill (C1)`: structured residual variance explained by spatial dependence.
- `Range (a)`: model scale of the residual variogram. Its relation to the practical range depends on the model, as shown in Section 11.4.

The residual semivariogram panel helps verify whether the RF residuals still show spatial dependence. If the residuals have a meaningful spatial pattern, RK can improve the final map by adding a kriged residual surface to the RF trend prediction. If the residual structure is weak, the RK map may be very similar to the RF map.

Information

RK should be interpreted as a hybrid method rather than a purely machine learning method. The RF stage explains variation using coordinates and covariates, while the kriging stage models spatial autocorrelation remaining in the residuals. Its performance therefore depends on both predictor quality and the quality of the residual variogram.

12.2.4 Adjusting the Residual Semivariogram

After fitting the RF trend with `Fit RF parameters`, click `Adjust semivariogram` in the residual panel. The popup exposes maximum distance, lag distance or number of lags, and the candidate models without separating the RF and kriging workspace into two windows.

Click `Update preview` after changing distance or lag settings. Click `Validate models` to compare the selected Spherical, Exponential, and Gaussian candidates. In this residual adjustment workflow the selection uses the lowest RMSE, with SSE used to resolve a tie. Inspect the numerical table and select a row before `Apply selected model`. The rule is displayed in the window; Geostatistics prioritizes LCCC, while the Framework semivariogram popup prioritizes R^2.

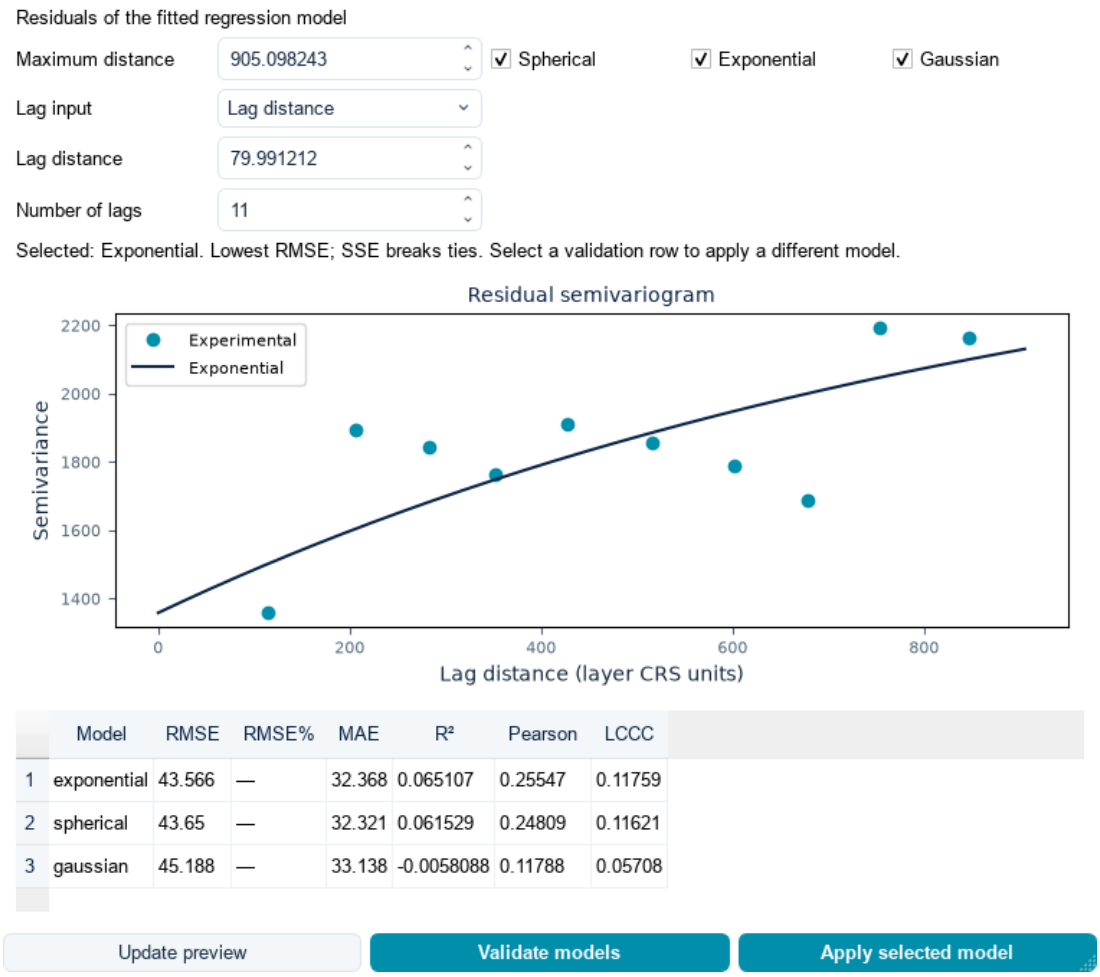


Figure 16: Residual semivariogram adjustment: distance and lag controls, model preview, numerical validation, and Apply selected model.

Return to the RK workspace and click `Interpolate` to generate the sum of the RF trend

and kriged residual prediction. Run RK validation from its `Validation` subtab. After changing the predictors, RF fit, or residual settings, recalculate the dependent stages rather than accepting an earlier result.

13 Prediction and Uncertainty Maps

After interpolation, the map selector provides the prediction and the available diagnostic components. Geostatistics labels this selector `Map`; RF, RK, and Framework use `Map output`. Standard deviations (SD) use the response units, and variances use squared response units. The meaning of an SD depends on the method; the maps below are not interchangeable measures of predictive error.

13.1 Conditional Kriging Uncertainty

Ordinary and linear Universal Kriging produce a prediction, `Kriging standard deviation`, and `Kriging variance`. The variance is the conditional kriging prediction-error variance at point support, and SD is its square root. It includes estimation of the unknown mean or linear trend but treats the fitted covariance parameters and drift specification as fixed. It does not account for variogram-parameter uncertainty or all sources of data and model error.

The implementation treats sampled observations as exact: the semivariogram is zero at zero separation, while the nugget contributes at positive separation. A prediction at an observed sample location therefore reproduces the observation, with conditional variance zero there up to numerical precision. Measurement error is not modeled separately. A positive nugget does not turn these outputs into uncertainty for a noisy observation or a future measurement. Interpret SD with the fitted model, sampling geometry, and validation results; it is not a calibrated prediction interval.

13.2 Random Forest Tree Dispersion

RF adds `RF tree standard deviation` and `RF tree variance`. At a prediction location, these summarize disagreement among the individual fitted trees. With T trees and tree predictions $\hat{z}_t$, the sample variance is

$$s^2_{\text{tree}} = \frac{1}{T-1}\sum_{t=1}^{T}(\hat{z}_t - \bar{\hat{z}})^2, \qquad s_{\text{tree}} = \sqrt{s^2_{\text{tree}}}.$$

At least two trees are required. Tree SD is not the standard error of the forest mean or a calibrated prediction interval. It excludes outcome noise and model bias; a low or zero value does not demonstrate that a prediction is accurate. Use it as a diagnostic alongside validation and predictor coverage. The forest prediction continues to use the fitted model's usual prediction routine.

13.3 Regression Kriging Components

RK provides `Residual kriging standard deviation`, `Residual kriging variance`, `RF trend tree standard deviation`, and `RF trend tree variance`, in addition to the final RK prediction. Residual kriging SD is conditional uncertainty for kriging of the RF residuals, holding the fitted RF trend and residual variogram fixed. It excludes uncertainty in the RF trend, variogram parameters, and covariates. The RF trend tree outputs describe tree disagreement separately, using the RF definition above.

These components are not combined into a total RK predictive variance. Do not add the RF tree variance to residual kriging variance or interpret either component as a calibrated interval for the final RK prediction. They describe different aspects of the fitted workflow.

13.4 Displaying, Adding, and Saving Outputs

1. Before interpolation, set `Add SD layer`. It is checked by default in Geostatistics, RF, and RK. It controls whether the next run adds the primary SD layer to the QGIS layer tree and map canvas, alongside the prediction. The primary SD is kriging SD for OK/UK, tree SD for RF, and residual kriging SD for RK.

2. Run interpolation. The prediction is added to QGIS. Available SD and variance rasters are calculated even when `Add SD layer` is unchecked; the checkbox controls addition to QGIS, not their calculation. Variance and RK trend-dispersion layers are not added automatically.

3. Use the map selector to inspect each available output in the plugin. Prediction and diagnostic maps retain independent palette, value limits, and view settings. Switching the preview does not rerun the model or change its prediction values.

4. Preserve outputs using the export setting in `Data`. Prediction and component GeoTIFFs share the prediction grid, CRS, extent, and NoData mask. With temporary outputs, export required files before closing QGIS. Unchecking `Add SD layer` after a run does not remove a layer already added.

Framework has its own `Add SD layer` choice, independent of the standalone tabs. It is enabled for selected OK, RF/RFE, and RK methods. Diagnostic choices become available after a successful matching Framework interpolation. IDW, TPS, and SVM do not provide uncertainty maps in v2.1. The Framework prediction remains the prediction shown in the report when another diagnostic map is being previewed; available uncertainty components appear in their own report section.

14 Validation of the Selected Interpolator

Validation is used to evaluate the result produced by the interpolator selected by the user. In the plugin, the validation tabs follow the same general structure across deterministic, geostatistical, and machine learning methods, but each validation result refers to the method currently being evaluated. The user can run validation from the corresponding method tab after the common inputs and method-specific parameters have been defined.

Figure 17 shows the common validation layout. The observed-versus-predicted plot displays the validation points, the 1:1 reference line, and the fitted trend line. The metrics panel reports the numerical validation results, while the controls below it allow the user to choose between automatic CV, LOOCV, and K-Fold.

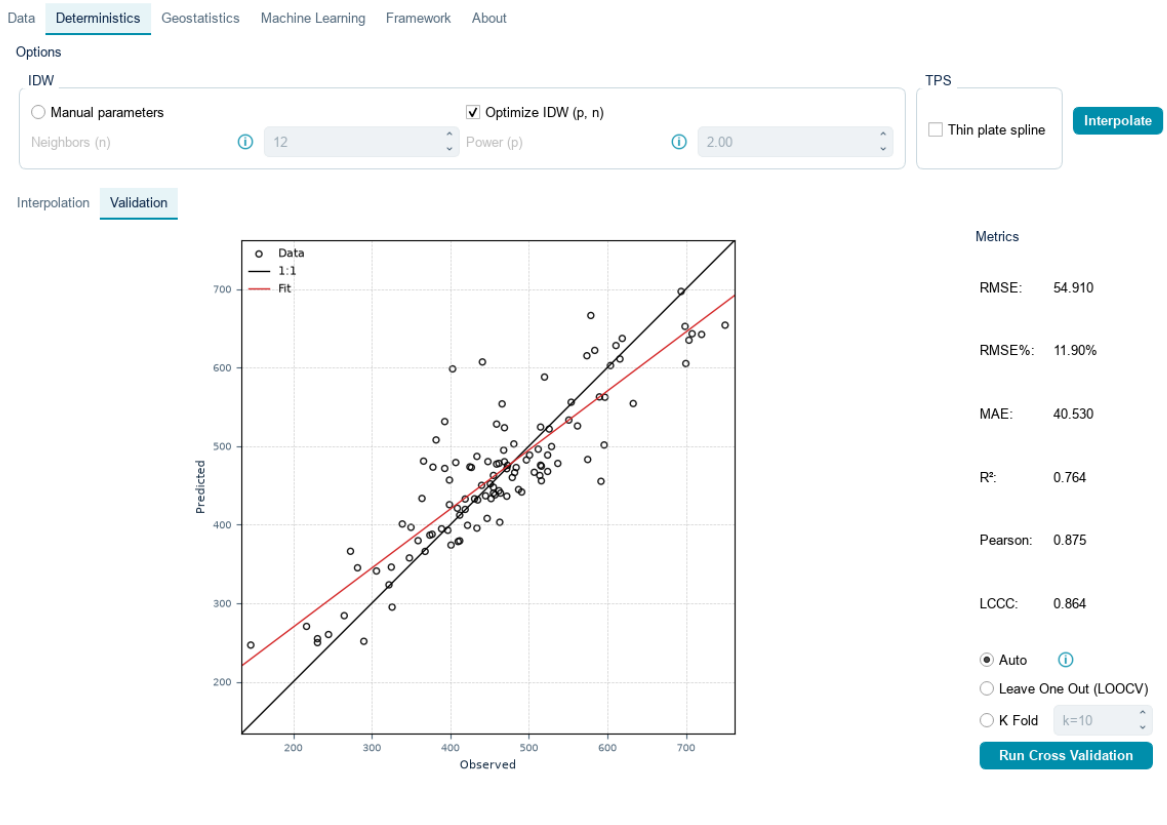


Figure 17: Validation tab showing the observed-versus-predicted plot, validation metrics, and cross-validation mode controls.

The available validation modes are:

1. **Auto.** The plugin selects the validation strategy according to the number of valid observations used by the method. If the dataset has $n \leq 100$ valid observations, the plugin uses Leave-One-Out Cross-Validation (LOOCV). If the dataset has $101 \leq n \leq 500$, the plugin uses K-Fold cross-validation with $k = 10$. If the dataset has more than 500 valid observations, the plugin uses K-Fold cross-validation with $k = 5$. This is the general automatic CV rule. Massive-data validation uses the disclosed bounded hold-out workflow described in Section 19.2.

2. **Leave-One-Out Cross-Validation (LOOCV).** The plugin removes one observation at a time, fits or predicts using the remaining observations, and validates the prediction at the omitted location. LOOCV uses all possible single-point omissions and is useful for small datasets, but it can be slow when the number of observations is large.

3. **K-Fold Cross-Validation.** The plugin randomly divides the observations into k groups. For each fold, one group is used for validation and the remaining groups are used for fitting or prediction. K-Fold is faster than LOOCV for medium and large datasets because it requires fewer validation runs.

The value of n refers to the valid observations available to the method at validation time. For machine learning methods, this means observations with the target variable and complete values for the selected predictors. If missing covariate values are present, the number of valid observations used for validation may be smaller than the original point-layer count.

Validation outputs are reported through metrics and observed-versus-predicted plots, as illustrated in Figure 17. The metrics evaluate the selected interpolation run and may include RMSE, RMSE%, MAE, R^2, Pearson correlation, and LCCC, depending on the method tab. Lower error metrics indicate smaller prediction errors, while higher agreement or correlation metrics indicate stronger agreement between observed and predicted values.

Information

Validation evaluates how the selected interpolator predicts values at sampled locations, not how visually smooth or detailed the output map appears. A visually detailed map is not necessarily more reliable, so the validation metrics and observed-versus-predicted plot should be checked before accepting the interpolation result.

15 Framework-Guided Interpolation

The Framework tab is not an additional interpolation model. It is a decision-support component that organizes the information already loaded in the plugin and guides the choice of an interpolation route. This framework is based on the scientific criteria described by DELGADO BEJARANO et al. (2026), so the recommendation is not arbitrary; it follows data characteristics and spatial evidence. The plugin uses two possible framework paths, depending on whether auxiliary covariate rasters are available.

15.1 Framework Overview and Data

The first window in the Framework tab is an overview of the current dataset. This screen brings together the information needed before choosing the framework path. The target variable, pixel size, and number of samples come from the selections made in the Data tab. Moran's I, the p-value, the spatial pattern, the Spatial Dependence Index (SDI), and the SDI status summarize the spatial diagnosis used by the framework. The semivariogram preview is calculated automatically from the current data using the active Geostatistics semivariogram configuration. It is synchronized as soon as the Framework preview is displayed, and the same configuration can be edited through *Semivariogram settings*.

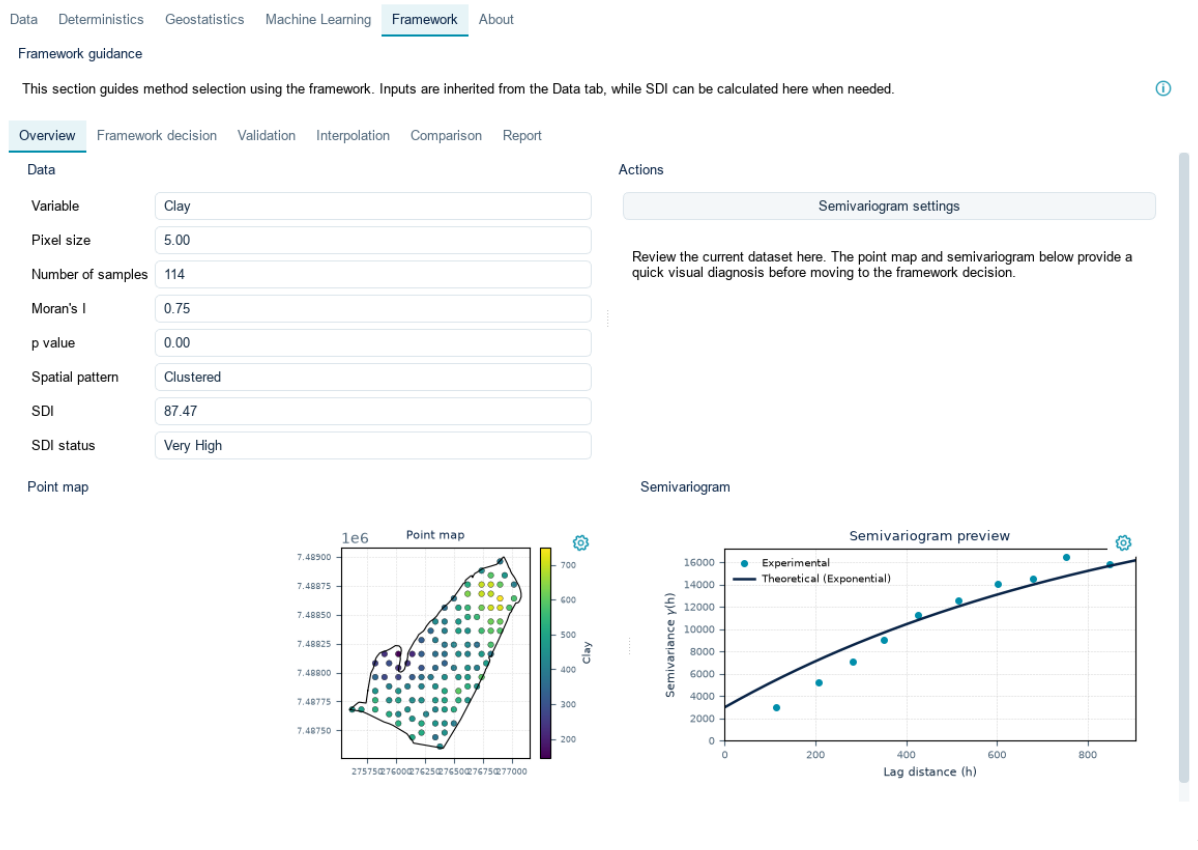


Figure 18: Framework overview showing the dataset information, point map, and semivariogram preview used before choosing the framework path.

The point map and semivariogram preview provide a visual check before moving to the *Framework decision* tab. The point map helps verify the sampled values and their distribution within the boundary polygon. The semivariogram preview helps the user identify whether there is spatial structure that may justify a geostatistical route. If the automatic semivariogram does not represent the dataset adequately, the user can adjust the model settings before continuing. At this stage, the user is reviewing the dataset and the spatial evidence that will support the framework decision.

15.2 Framework Decision Paths

After the overview has been checked, the user selects the framework mode in the *Framework decision* tab, as shown in Figure 19. The plugin provides two decision paths:

- **Univariate framework**: used when the interpolation is based only on the sampled point layer, the target variable, the boundary polygon, the pixel size, and the spatial diagnostics. Covariate rasters are not used in this route. The available methods are TPS, OK, and IDW.
- **Full framework**: used when auxiliary raster covariates are available. When this mode is selected, the *View covariates* option opens the same covariate-preparation window mentioned in Section 12.1 and shown in Figure 12. The available methods are TPS, OK, IDW, SVM, RF, and RK.

The framework figure is dynamic. It is updated according to the characteristics of the current dataset, including sample size, spatial pattern, Moran's I significance, SDI, and the selected framework mode. After the user clicks *Evaluate available methods*, the plugin follows the selected framework path and displays the suggested method in the decision summary and suggested-methods panel.

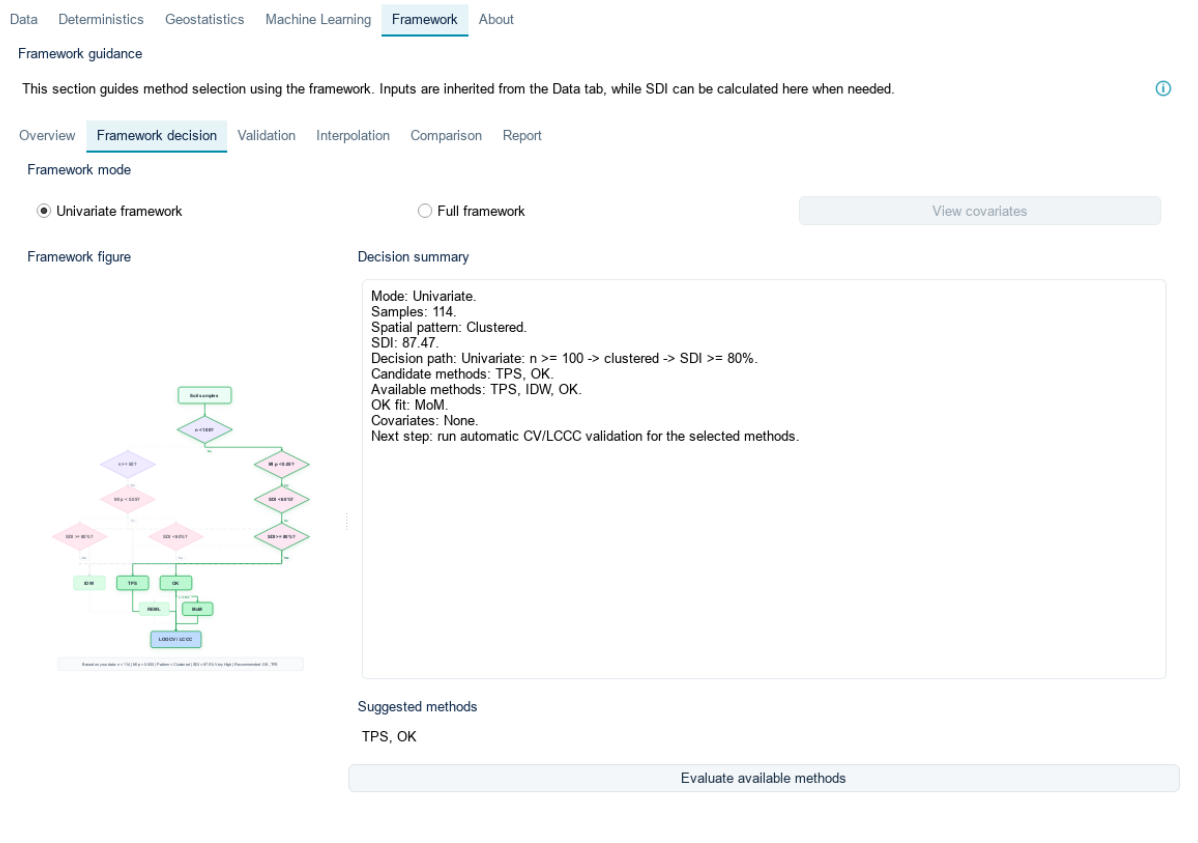


Figure 19: Framework decision tab showing the framework mode selection, the univariate decision summary, and the suggested method for the current dataset.

Information

The overview screen is common to the framework workflow. It helps the user review the data, Moran's I, SDI, point distribution, and semivariogram before deciding whether the recommendation will follow the univariate route or the full framework with covariates.

15.3 Framework Validation

The Validation tab evaluates the methods selected within the framework workflow. After the framework decision has been generated, the plugin can automatically select the recommended methods through *Select suggested*. The user can also manually activate additional methods before running validation. This is useful because the framework recommendation is a technical guide based on general data characteristics, not an absolute rule. For a given set of plugin settings, interpolation performance is strongly data-dependent and is affected by the structure, distribution, variability, and quality of the current dataset. Therefore, the framework should be interpreted as a decision-support tool that narrows the set of reasonable methods, while validation provides the empirical evidence for the selected data.

When the user clicks *Run validation*, the plugin validates the selected methods and reports the results in a table. The table includes error, agreement, and correlation metrics such as RMSE, RMSE%, MAE, R^2, Pearson correlation, and LCCC. The validation summary identifies the winner among the methods included in that validation run. The Framework ranks methods by the highest LCCC and then the lowest RMSE; this method ranking differs from the semivariogram model ranking.

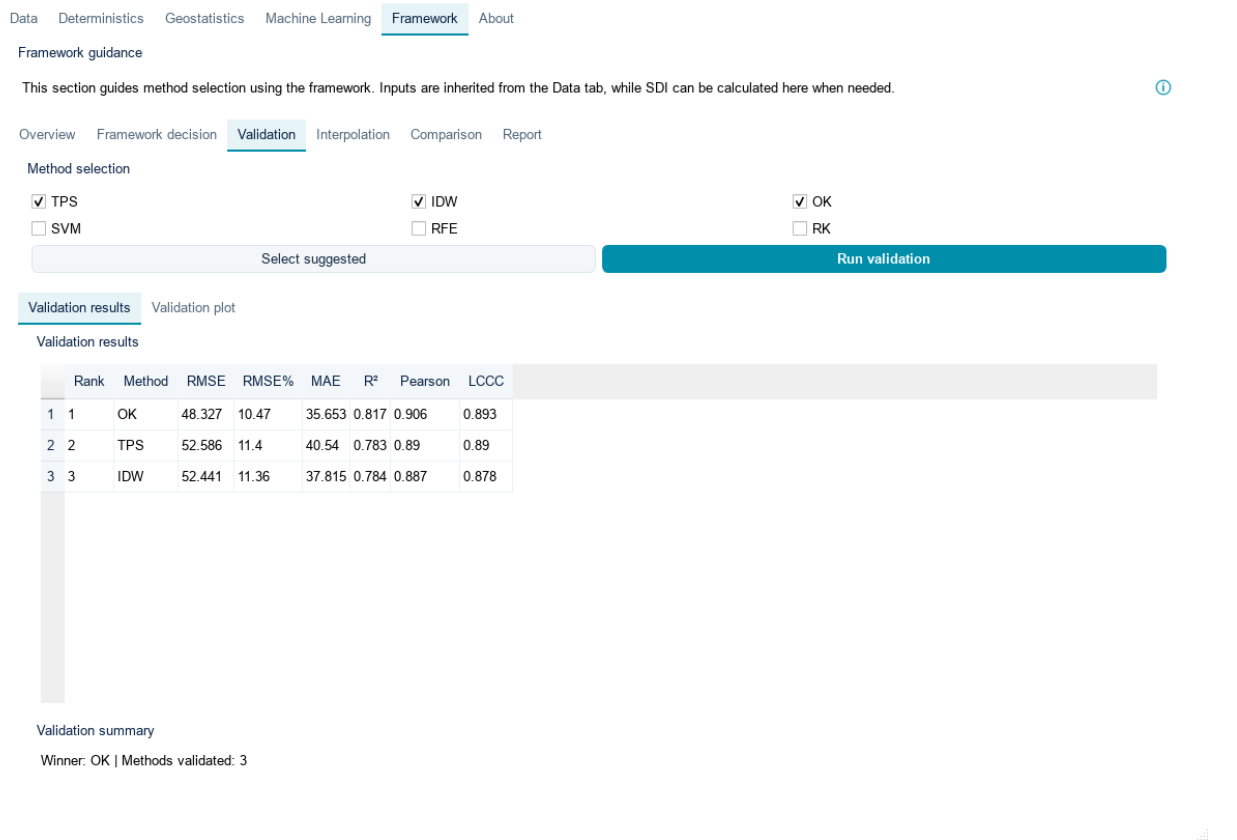


Figure 20: Framework validation tab showing method selection, the validation results table, and the validation summary for the selected methods.

The Validation plot tab provides graphical support for interpreting the results. The user can compare the selected methods through LCCC and RMSE plots. Higher LCCC indicates stronger agreement between observed and predicted values, while lower RMSE indicates smaller prediction error. In addition, the user can inspect observed-versus-predicted plots for each selected method,

which helps identify bias, dispersion, or systematic under- and over-prediction.

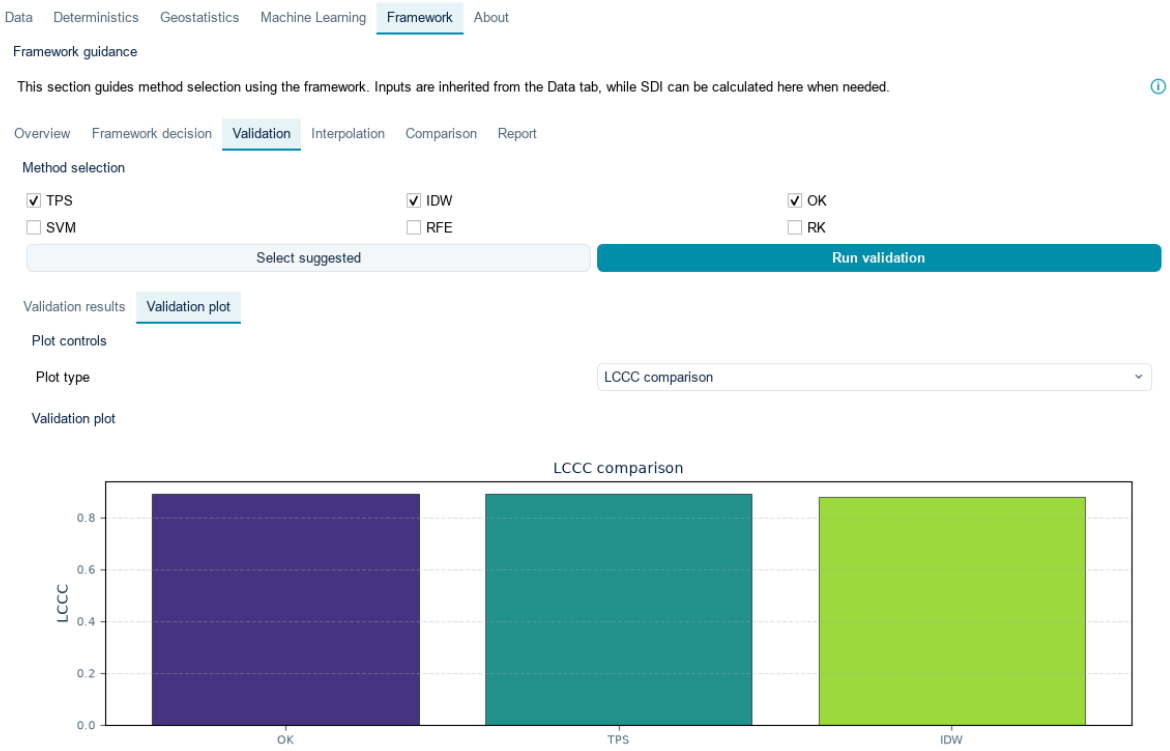


Figure 21: Framework validation plot tab showing graphical comparison of the selected methods using LCCC; RMSE comparison and observed-versus-predicted plots can also be selected.

Information

The method suggested by the framework is the method expected to be suitable according to the framework criteria, but the final decision should also consider the validation metrics and plots obtained for the current dataset.

15.3.1 Automatic Machine Learning Search in Framework

Framework automatically tunes RF, SVM, and the RF trend of RK. It uses the same effective default ranges, candidate construction, and bounded search routines as the individual windows. Opening or configuring a standalone method is not required, and a standalone `Manual parameters` choice does not disable Framework search. RF may be labeled `RFE` in the original Framework diagrams; both labels refer to Random Forest here.

Click `ML grid search` in Framework Validation to review or change each method's ranges, search folds, and maximum sampled combinations. `Restore Defaults` changes the dialog controls; accept to apply them or cancel to discard edits. The default budgets are three search folds with at most 10 combinations for RF, 12 for SVM, and 10 for the RK trend. Dense and massive profiles can reduce these budgets as described in Section 19.

The control name `Grid search` denotes reproducible sampling from a finite candidate grid, bounded by the iteration limit, rather than an exhaustive evaluation of every possible combination. RF and RK tuning prioritize search-CV MAE; SVM tuning prioritizes search-CV RMSE. The selected model is then fitted on all eligible training rows for that run. Inspect the report for the effective candidate values and selected parameters.

A completed search can be reused within the same plugin session only when its training values, predictor order, target, and search configuration match. The final model is fitted again; reuse and the original search record are identified in the report. Changed inputs or search settings require a new

search. Standard performance-validation scores follow parameter selection and are not a nested-CV estimate. Automatic CV and K-Fold do not create separated spatial validation blocks; spatial block validation is not implemented in v2.1.

15.4 Final Decision and Interpolation

The `Interpolation` page uses the best validated method by default while `Use best validated method` is active. Disable this option to select another available validated method. Changing the selection prepares the next run; it does not change a raster already generated.

Click `Run interpolation` to generate the final Framework map. Its canvas remains empty until a Framework interpolation succeeds; standalone maps and validation alone do not populate it. Framework interpolation, validation, and comparison use private result workspaces, so they do not populate or overwrite maps, models, parameter values, or validation results in the individual tabs. Standalone reruns likewise do not replace a completed Framework map. Shared input changes still invalidate results that no longer match the data.

The summary distinguishes the initial framework recommendation, validation winner, selection for the next run, and method actually interpolated. The Report describes the executed interpolation. For example, choosing IDW after running TPS leaves the report's executed method as TPS until a new IDW interpolation finishes successfully. Set Framework's own `Add SD layer` option before running a supported method; see Section 13.

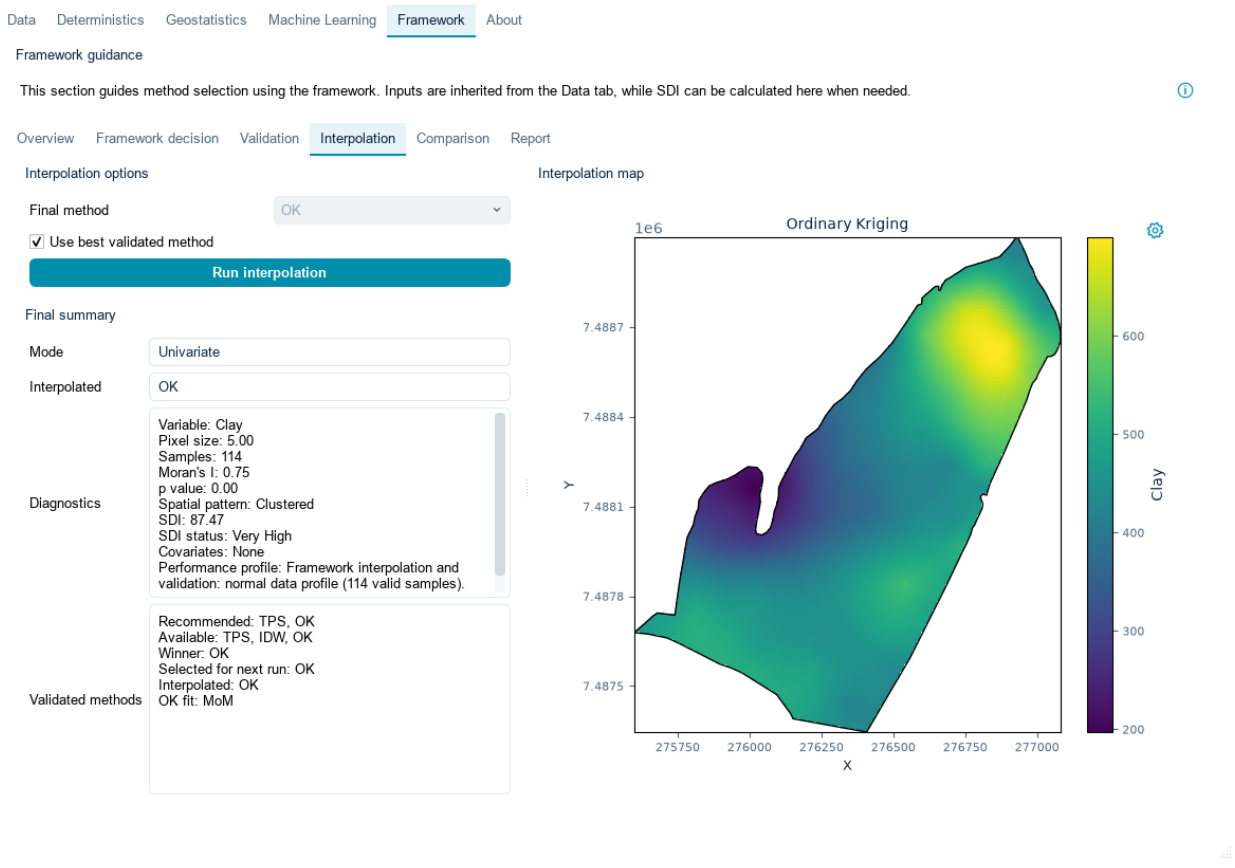


Figure 22: Framework interpolation page: final method selection, map, and executed-result summary.

15.5 Validating Models in Framework Semivariogram Settings

From `Overview`, click `Semivariogram settings`. Review the fitting method, maximum distance, lag distance or lag count, model, nugget, partial sill, and range. `MoM` shows experimental semivariances and a theoretical curve; `REML` shows the theoretical curve only. Automatic MoM uses the joint constrained weighted fit described in Geostatistics. `Range (a)` is the model scale, with the model-specific practical range defined in Section 11.4.

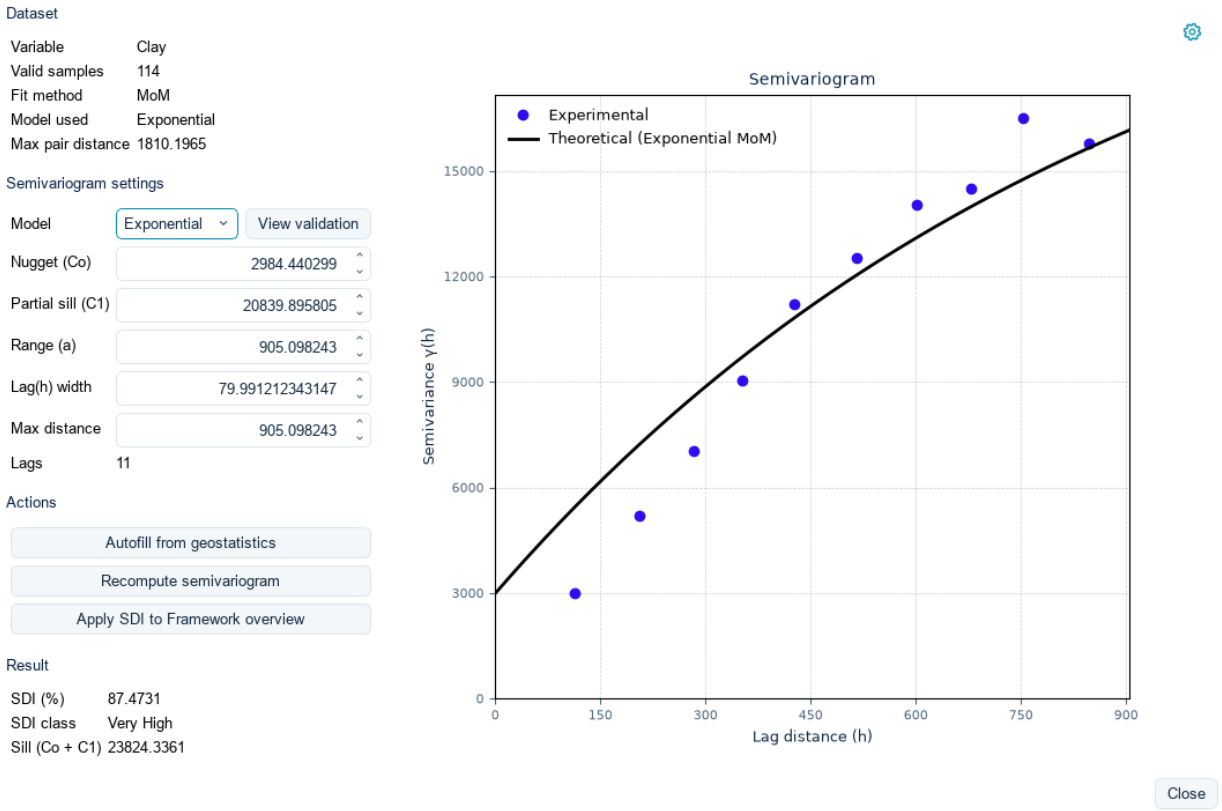


Figure 23: Framework semivariogram settings with model validation beside the current distance, lag, and parameter controls.

Click `View validation` to compare Spherical, Exponential, and Gaussian models with the current data and settings. The table reports cross-validation metrics and explains the automatically recommended model: highest R^2, then lowest RMSE. Select a successful candidate row and click `Apply selected model` to use its fitted parameters. Review the new SDI and apply it to the Framework overview when appropriate.

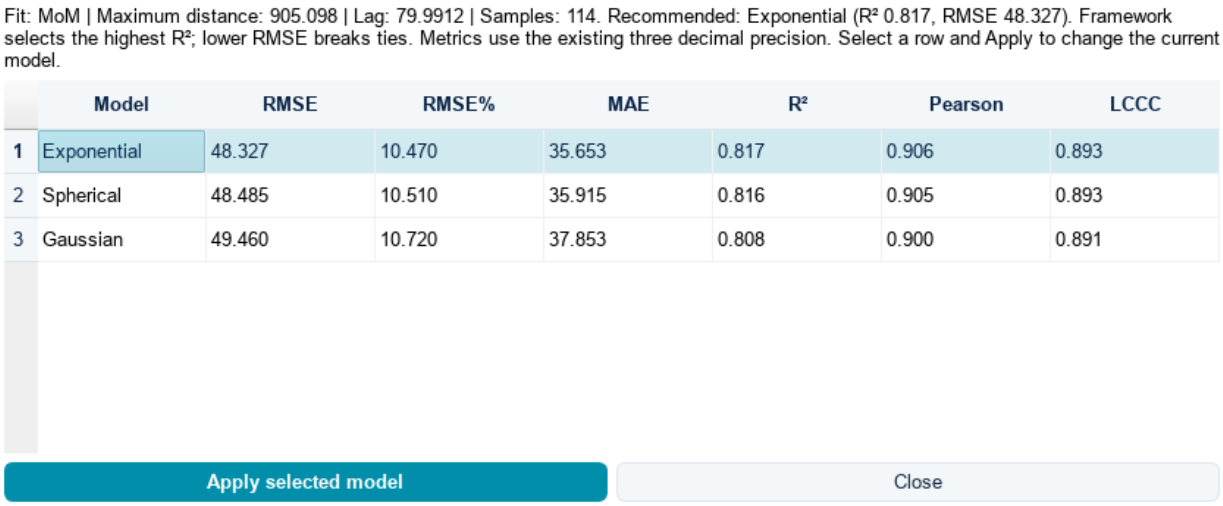


Figure 24: Framework model validation: all three candidates, ranking explanation, and Apply selected model.

If the data or settings change while validating, reopen validation for the current configuration. An earlier table should not be used to apply stale parameters.

16 Comparison of Maps

The Framework `Comparison` page compares maps. Quantitative method-validation results remain in `Validation`. Use both views: validation assesses prediction error, whereas map comparison shows where spatial predictions differ.

1. Validate the candidate methods using the current input data. In Full mode, prepare covariates before validating RF, SVM, or RK.
2. Open `Comparison`. Choose two different available methods in `Map A` and `Map B`. Validated alternatives are listed even if their maps have not yet been generated.
3. Click `Compare maps`. The plugin generates missing candidate maps on the current grid, then displays A, B, and their signed difference. Comparison does not replace the executed final Framework result.
4. Read the difference as $D = A - B$. Positive cells mean A predicts a higher value; negative cells mean B predicts a higher value. The difference uses the target variable's units. Reversing A and B reverses the sign.
5. Click the corner configuration button to use a common palette, value range, and extent. Compare A and B with a shared value range so that equal colors represent equal values. The difference has its own diverging color scale.
6. Pan or zoom to inspect the same area in both maps. Use `Fit to layer` to return to the full extent.
7. Enable `Include this comparison in report` after producing the comparison you want to document.

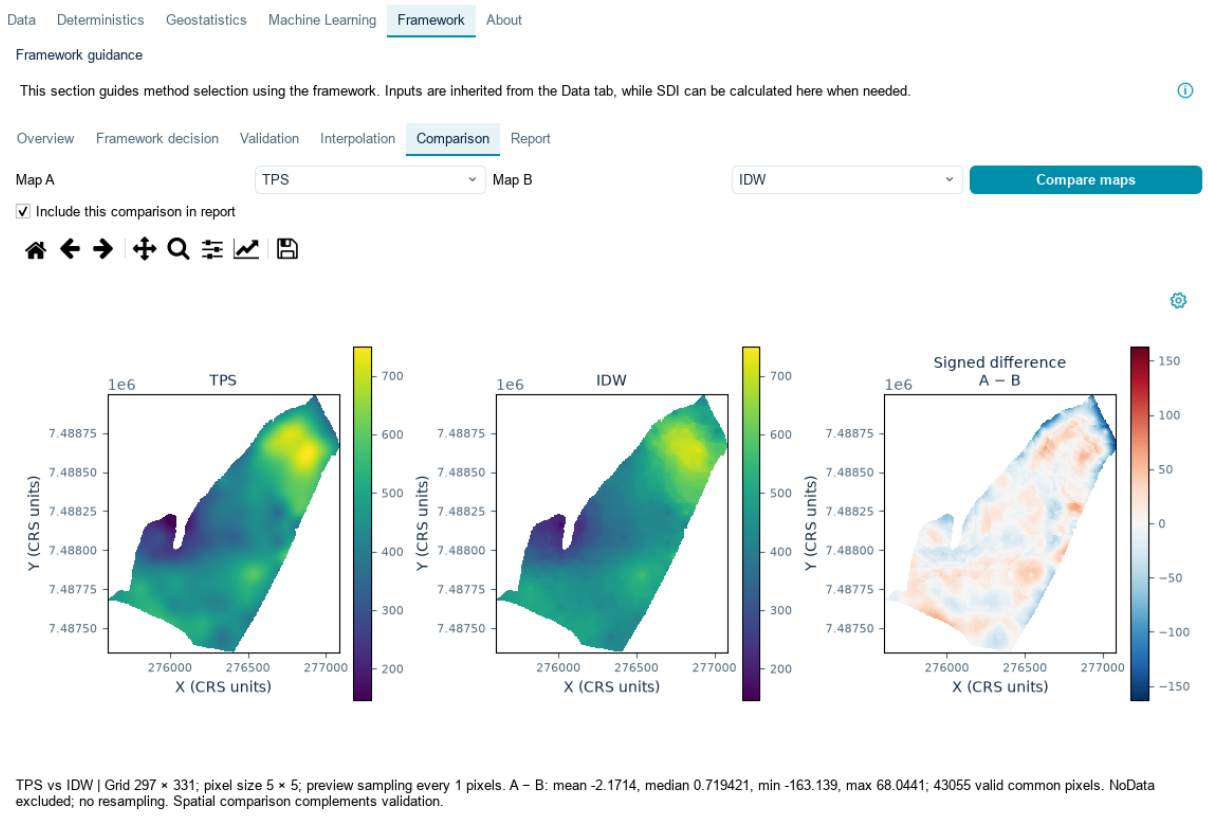


Figure 25: Comparison page with Map A, Map B, and the signed A minus B map on the common raster grid.

The rasters must have the same CRS, grid dimensions, geotransform, pixel size, and alignment. The calculation excludes pixels with NoData in either map and does not resample incompatible grids. Difference statistics use all valid common pixels; large-grid previews may display a sampled rendering, which is disclosed in the status line. A small mean difference can hide large positive and negative local differences, so inspect the map, range, and distribution as well.

If a candidate map cannot be generated, inspect the status message and correct its prerequisites, such as missing predictors or a failed model. The previously executed final map remains recorded. Changing the candidate selection invalidates the earlier comparison and requires `Compare maps` again.

17 Map Display and Larger View

Spatial maps have a configuration button in the plot corner. It opens a popup with visible color gradients, palette selection, automatic or manual value limits, numeric scale, and `Fit to layer`. These controls change the display, not the predictions or validation metrics. Numeric map scale requires a projected CRS and uses its map units.

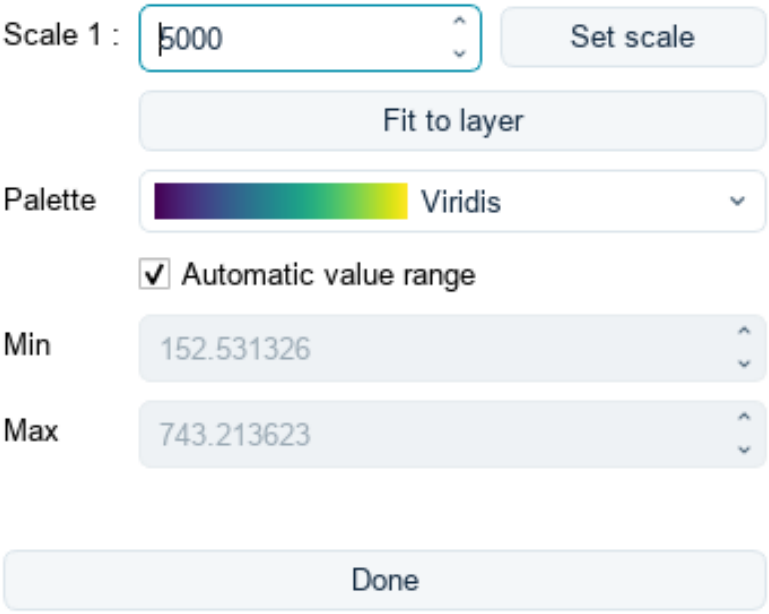


Figure 26: Map display settings: scale, fit, visible palette preview, and value range.

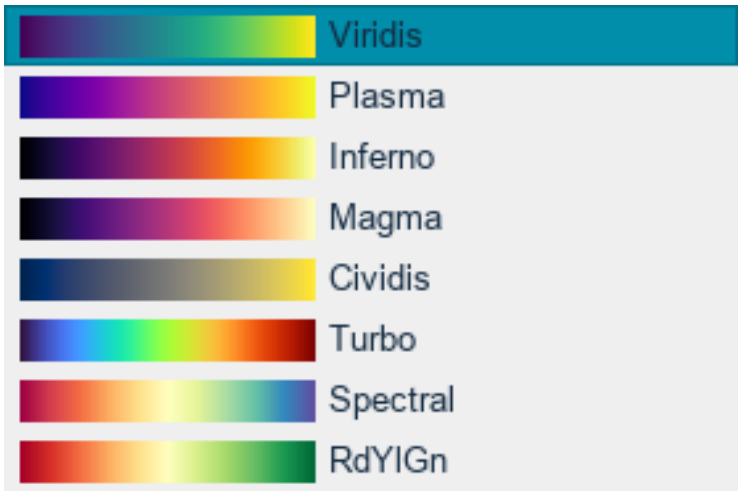


Figure 27: The palette selector shows color gradients as well as palette names.

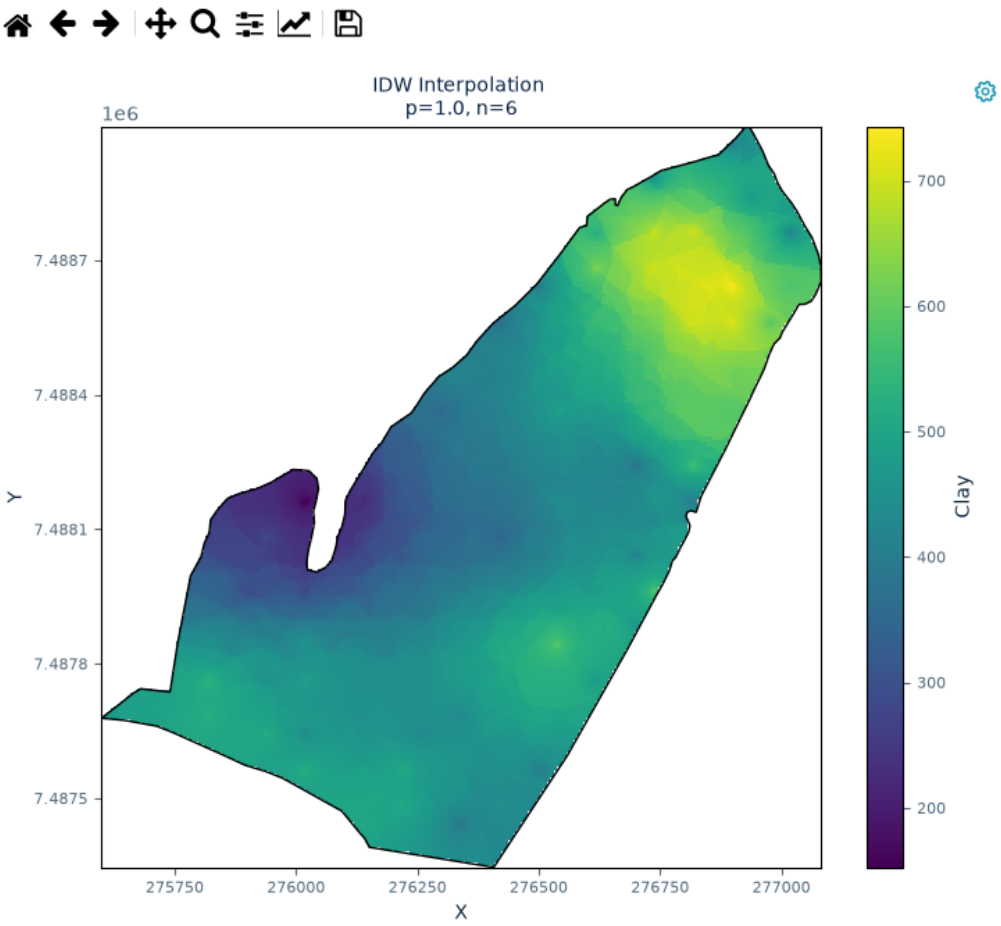


Figure 28: Enlarged map view retaining the current styling and its corner configuration button.

Open `View larger view` from the plot context menu when a larger map is useful. The enlarged view preserves the map data, NoData mask, palette, value range, and current extent. Its corner configuration button opens the same display settings. Keep a common scale and color range when presenting comparable maps. Validation plots retain fixed scientific colors and do not offer a map-palette selector.

18 Interactive Reports and Export

The Framework `Report` page provides an interactive summary of the current session. Use the section list to review the overview, validation, interpolation, and diagnostics available for that session. The initial recommended method set is reported separately from the validation winner, next selected method, and executed interpolation. Links return to the relevant analysis pages.

The report records completed ML searches and selected parameters, including the effective search settings, folds, criterion, score, evaluated combinations, and training population. Reused searches retain their original provenance; later control changes do not rewrite an earlier execution. Available uncertainty components and their definitions are reported separately from prediction maps. When a map comparison is included, its full A/B/signed-difference figure is kept together. Newly generated report text follows the selected interface language.

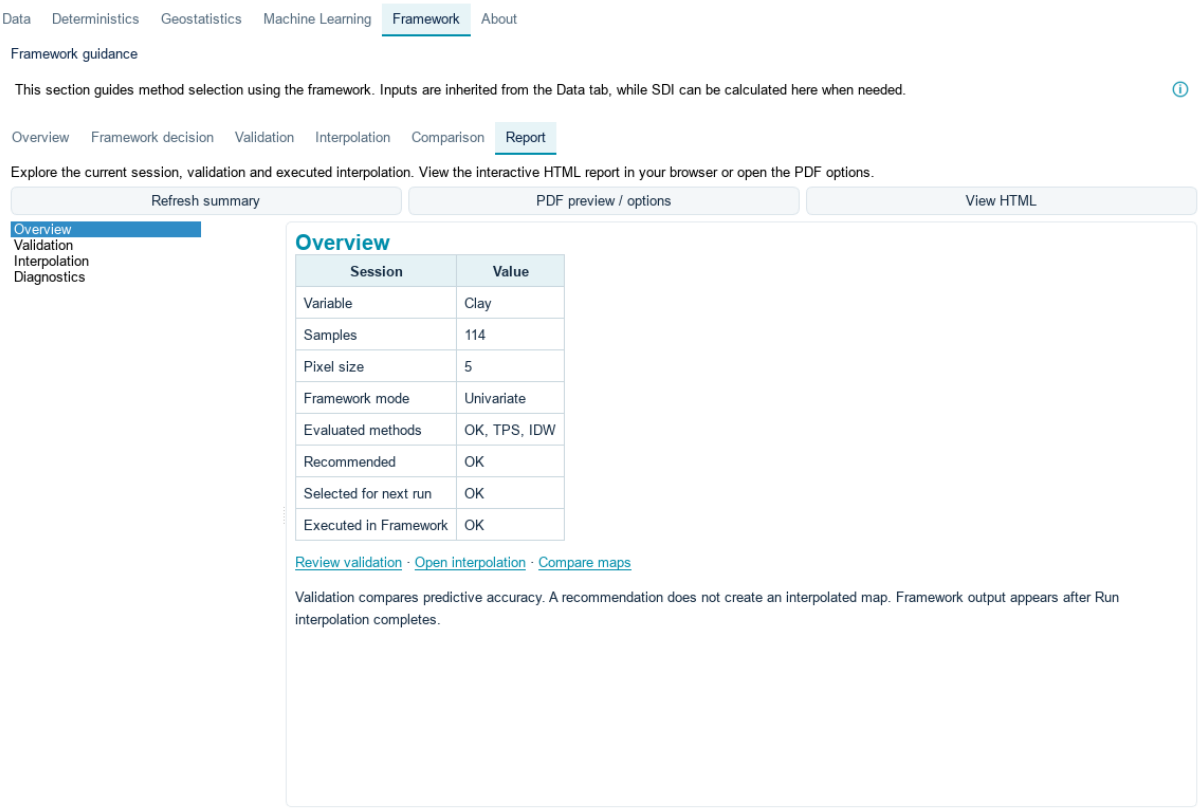


Figure 29: Report page: interactive session summary with PDF options and View HTML.

18.1 Temporary HTML Preview

Click `View HTML` to open a temporary HTML report in the browser. No destination filename is required and the user does not need to download the HTML first. The report embeds its figures, provides navigation links to its sections, and lets the user expand or collapse sections individually or with `Expand all` and `Collapse all`. This temporary preview is convenient for reviewing the results during the session; save or export the material if it must be kept permanently.

18.2 PDF and Raster Outputs

Click `PDF preview / options` to choose report content, inspect the paginated preview, and export a PDF. Available content depends on completed analyses and can include the point map, semi-variogram, covariates, validation metrics and plots, framework decision, final interpolation, diagnostic results, and the included map comparison. A report can only document analyses that have actually been run.

Raster persistence is controlled by `Export Rasters to project folder` in `Data`. Enable it before interpolation to write the raster in the project output directory. If it is disabled, preserve a required temporary layer through QGIS export before closing the application. Keep the raster, report, parameter settings, data exclusions, and software environment together to document a mapping decision.

19 Dense and Massive Datasets

The processing profile is selected automatically from the number of valid observations available to the current method. Density here names a computational profile based on sample count; it is not a physical sampling-density measurement. Missing values, missing predictors, and explicit exclusions may make this count smaller than the source layer's feature count.

Table 9: Automatic processing profiles. Boundary counts belong to the lower profile.

Profile	Valid observations	Main implication
Normal	$n \leq 500$	Original small-sample processing paths
Dense	$500 < n \leq 10{,}000$	Bounded tuning, local spatial prediction, and sampled variogram pairs
Massive	$n > 10{,}000$	Smaller prediction blocks and a bounded validation hold-out

There is no fixed stop at 100,000 observations. Productivity and sensor datasets around this scale and above can use the massive profile. Practical capacity depends on memory, output grid cells, covariates, neighborhood size, and the method. The 50,000-observation computational notice is advisory rather than a hard ceiling. A 32-bit QGIS process has a smaller address space than a 64-bit installation.

19.1 What Changes with the Profile

- **Machine learning tuning.** Dense search limits the requested tuning to at most three folds and eight parameter candidates. Hyperparameter search uses a reproducible subset of at most 2,000 valid rows where needed. This search subset is separate from the final fit, which uses all valid rows available to that model.
- **Random Forest.** In massive mode, the resolved forest is bounded to at most 300 trees and a minimum terminal-node size of 5, including manually requested settings. Review the resolved parameters shown by the run.
- **SVM.** Above 500 observations, the scalable path uses an approximate kernel representation rather than the small-sample exact radial-basis SVR path. It uses up to 256 Nystroem components for dense data and 128 for massive data. Interpret this as a computational approximation and validate its suitability.
- **Ordinary and residual kriging.** Dense prediction uses local systems with a maximum neighborhood of 64 samples and a minimum of 16 where the available neighborhood permits. MoM replaces expensive REML fitting for large sample counts.
- **TPS.** Dense and massive profiles use local neighborhoods of 64 and 32 observations, respectively. This is a change from the global spline used with normal datasets.
- **Semivariogram.** Dense fitting bounds the sampled pair budget to 250,000 and proposes at most 36 lag classes; massive mode proposes at most 24. The displayed fit records the data profile. Pair sampling limits the calculation rather than deleting source observations.
- **Prediction and plotting.** Large predictions are processed in blocks; the shared block sizes are 25,000 rows for dense data and 5,000 for massive data where used. Point previews may use up to 10,000 representative samples; plots above 50,000 observations may use hexagonal density bins. A display sample does not define the final fitting population or raster statistics.

19.2 Validation for Large Data

The general Auto rule uses LOOCV up to 100 observations, 10 folds from 101 through 500, and 5 folds above 500. Above 10,000 valid observations, the Framework and machine learning validation paths use a reproducible hold-out of about 20%, capped at 10,000 test observations. The remaining rows train the validation model; the final interpolation model is fitted afterward using its full valid population.

In the Framework, held-out locations are selected to represent the spatial coverage; this is not the same as validation with separated spatial blocks or an independent field campaign. Standalone

RF, SVM, and RK use reproducible random hold-out indices. Read the actual strategy and test count in the result rather than describing these runs as full LOOCV. The automatic policy does not imply that every standalone method silently replaces a manually requested expensive validation.

19.3 Spatial Diagnostics at Large Scale

The automatic Global Moran calculation in `Data` uses bounded permutations to keep the initial overview responsive: up to 199 in dense mode, 49 above 10,000 observations, and 19 above 250,000. Fewer permutations make p-values coarser. The Outlier diagnostic window has its own explicit permutation choices (99, 499, 999) and local significance settings; those choices are not silently replaced by the automatic overview limits.

For very large inputs, start with KNN, a modest neighbor count, and 99 permutations to inspect the data. Increasing the distance radius can create very large neighborhoods and make local diagnostics expensive. Refine the settings when the first results justify the additional calculation, and record the actual settings used.

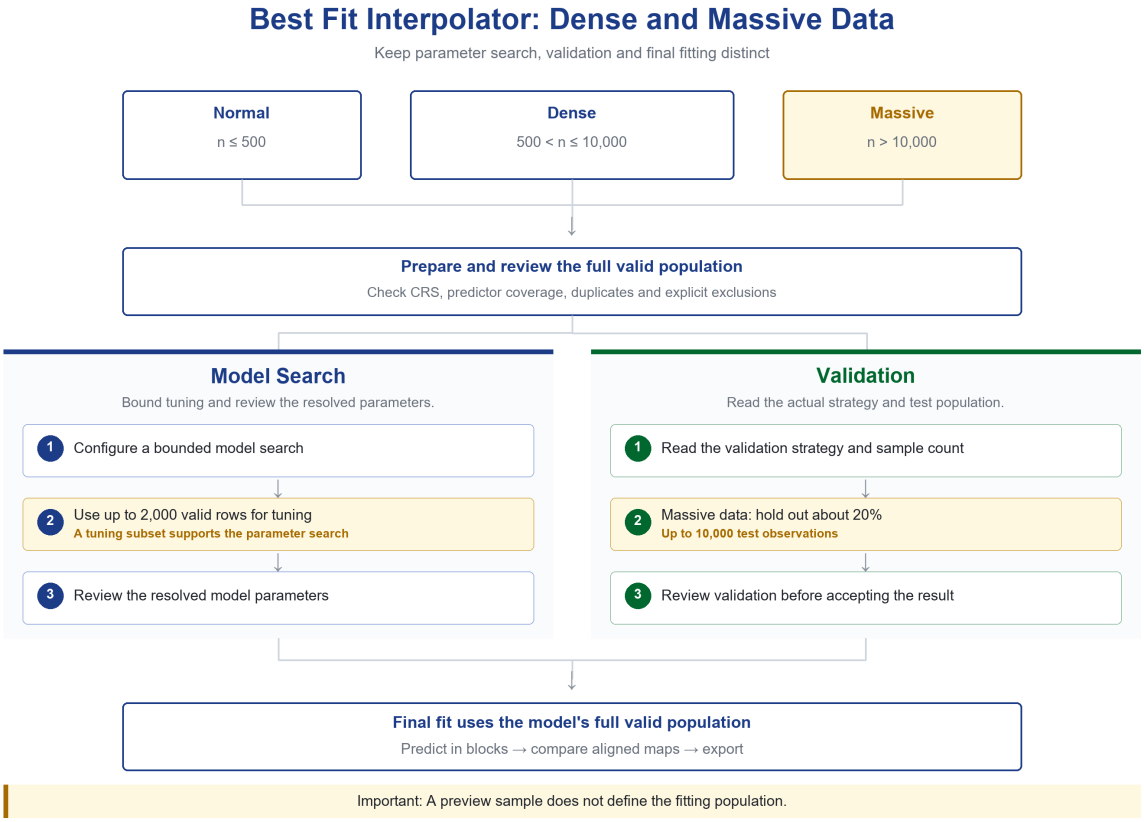


Figure 30: Processing flow for dense and massive data: distinguish tuning, validation, final fitting, and display samples.

19.4 Suggested Dense or Massive Workflow

1. Prepare the point values, projected CRS, boundary, and covariate coverage in QGIS. Remove duplicate coordinates before TPS and distinguish invalid measurements from legitimate extremes.

2. Load the data and read the reported valid sample count and profile. Review suspicious observations through Outlier diagnostic and record any explicit exclusion decisions.

3. Start with a practical pixel size. The number of grid cells scales approximately as mapped area divided by pixel area; halving the pixel width creates roughly four times as many cells.

4. Prepare only the predictors needed for the objective. Missing predictor coverage reduces the training population and may leave NoData in the output.

5. Evaluate candidate methods in Framework or fit one method manually. Review resolved tuning parameters, approximation notices, semivariogram settings, and the actual validation strategy.

6. Run the final interpolation, then compare two validated maps on the same grid. Inspect local differences and the validation metrics before accepting the result.

7. Review the interactive report and export the persistent raster and PDF needed for the project. Retain the source data, exclusions, parameter settings, profile, validation count, and software environment.

Information

The massive profile makes large computations more manageable; it does not guarantee a runtime or make fine pixels statistically more informative. A denser map is still limited by the observations and predictors supporting it. Independent or appropriately spatial validation may be needed when the aim is transfer to unsampled areas.

20 Practical Checks and Troubleshooting

Table 10: Common situations and checks.

Situation	Action
Missing or blank inputs	Load the correct layers in QGIS, choose a numeric target, and click Load. A new plugin session starts from clean inputs.
Map or distance mismatch	Check metric CRS, polygon validity, sample overlap, pixel size, and covariate alignment before interpolation.
Suspicious observation	Inspect the original measurement and its neighborhood; a statistical or spatial flag alone does not exclude it.
No experimental points in REML	Expected behavior: REML displays the fitted theoretical curve only. Use MoM when experimental semivariances are needed.
Changed variogram settings	Recalculate or update the preview, rerun model validation, and apply a model evaluated under the current settings.
Unavailable comparison	Validate both candidate methods and resolve missing covariates or failed predictions. Comparison requires two aligned rasters.
Report shows an earlier method	The report describes the successful executed interpolation. Changing the next-run selection requires Run interpolation before the executed method changes.
Dense run takes too long	Inspect grid-cell count, predictors, neighborhood, and requested validation; begin with a coarser grid and the automatic profile.
Missing ML dependency	Use the plugin's compatible dependency setup for the current QGIS Python and architecture. Do not copy compiled packages from another 32-bit or 64-bit installation.
Incomplete predictor coverage	Check the source raster paths, each raster CRS, extent, and explicit NoData settings. The run stops if fewer than half of eligible observations or inside-polygon grid cells have complete predictors; it does not fill missing values automatically.

Situation	Action
Map output or SD unavailable	Select OK, RF/RFE, or RK in Framework and complete its interpolation. For standalone Geostatistics, select OK or UK. IDW, TPS, and SVM have no uncertainty output.
Different range values	Distinguish the entered model scale from the practical range: a for spherical, $3a$ for exponential, and $\sqrt{3}a$ for Gaussian.
Language change delayed	Wait until the active calculation finishes. Regenerate a report to export it in the newly selected language.
Need to retain the session	Export the raster and report and record exclusions and settings before closing; reopening intentionally starts a clean session.

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